Homogeneous Three–Topology Comparison and Mode Dictionary in Einstein–Cartan + Nieh–Yan Theory: Geometric Structure of EC–Weyl Coupling

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Abstract

We systematically analyse the homogeneous minisuperspace of gravity with Einstein–Cartan (EC) connection and Weyl-squared coupling $\alpha C^2_{\rm EC}$ across three spatial topologies—$S^3\times S^1$, $T^3\times S^1$, and $\mathrm{Nil}^3\times S^1$—and determine how the EC–Weyl coupling modifies the mode dictionary of each topology. Extending the Levi–Civita framework of papers I–III to the EC connection $\omega_{\rm EC}=\Gamma_{\rm LC}+K$ (with K the contortion), we classify torsion modes as axial-scalar (AX), vector-trace (VT), and their product (MX), and establish nine theorems.

Theorem 1 (AX/VT dropout): In all three topologies, $C^2_{\rm EC}=C^2_{\rm LC}$ holds exactly in the AX ($V=0$) and VT ($\eta=0$) backgrounds; the EC correction appears only in the MX background as $C^2_{\rm EC}=C^2_{\rm LC}+16V^2\eta^2/(3r^2)$.

Theorem 2 (Palatini protection, EC): $G^{\rm EC}_{\eta\eta}=G^{\rm EC}_{VV}=0$ holds algebraically; torsion remains non-propagating and the spin-2 kinetic structure is unchanged from the LC case.

Theorem 3 (δ-sector null theorem): The auxiliary cubic coefficient $C_\delta:=\partial^3 V_{\rm eff}/\partial\delta_0\partial\delta_1\partial\delta_2|_{\delta=0}$ vanishes exactly in all three topologies and all torsion backgrounds, and $\partial_\alpha C_\delta=0$.

Theorem 4 (Palatini universality): In $S^3\times S^1$ the AX and VT spin-2/spin-1 mass spectra are identical: spin-2 mass $\mu^2=128\pi^2Lr/(3\kappa^2)-16384\pi^2L\alpha/(3r)$ (five-fold degenerate) and spin-1 mass $m^2_\omega=16\pi^2L^3/(\kappa^2 r)$ (three-fold degenerate).

Theorem 5 (torsion–volume coupling): In $T^3\times S^1$, $\partial^2V_{\rm eff}/\partial\varepsilon\partial s|_0=48\pi^4L\eta^2r/\kappa^2\neq 0$; the cross-term is of purely kinematic origin, independent of the Weyl mass.

Theorem 6 (Nil^3 EC slice minimum): For $\alpha<0$, $\mathrm{Nil}^3\times S^1$ supports a positive EC-induced slice minimum at $r_0=(4\kappa/\sqrt{3})\sqrt{|\alpha|}$ on the $\eta=V=0$ section; the full homogeneous Hessian is positive definite for $|\kappa^2\theta_{\rm NY}|<1$, marginal at equality, and a saddle above it. The scaling exponent is $\gamma=1/2$.

Theorem 7 (Nil^3 quintet splitting): The spin-2 quintet around the EC slice-minimum branch splits as $5 \to 0 + 2 + 1 + 1$; the zero mode q_3 is geometrically protected by the one-dimensional Lie structure of Nil^3.

Theorem 8 (uniaxial mass splitting): In the Nil^3 spin-1 sector, $m^2(\omega_2) \neq 0$ (non-commutative direction) while $m^2(\omega_0) = m^2(\omega_1) = 0$ (commutative directions).

Theorem 9 (ε-s cross-term classification): The origin of $\partial^2 V_{\text{eff}}/\partial\varepsilon\partial s$ differs essentially across topologies: purely kinematic for T^3, vanishing for S^3, and of pure curvature/Weyl origin for Nil^3.

These results establish a closed comparison framework for the homogeneous bulk geometric sector in EC+NY theory under the axial+vector-trace torsion ansatz.

1 Introduction

1.1 Background

In paper I [1] we compared $S^3 \times S^1$, $T^3 \times S^1$, and $\mathrm{Nil}^3 \times S^1$ within the Einstein–Cartan + Nieh–Yan (EC+NY) minisuperspace using a unified effective-potential framework, establishing a topology-dependent phase classification and demonstrating the energetic dominance of the isotropic $S^3 \times S^1$ vacuum.

In paper II [2] we focused on the isotropic $S^3 \times S^1$ vacuum and analysed in detail the effect of the Weyl-squared coupling αC^2 [7]. The Pontryagin density $P = \langle R, \star R \rangle$ [6] was shown to vanish identically under the homogeneous ansatz ("chiral equilibrium"), and conformal protection for $\alpha \leq 0$ together with asymptotic instability for $\alpha > 0$ were established, providing the physical basis for the geometric robustness of the S^3 vacuum.

In paper III (LC version) [3] we organised the complete set of homogeneous geometric degrees of freedom of $S^3 \times S^1$—spin-0 (radial r, torsion scalars η, V), spin-2 (traceless symmetric quintet q_A), and spin-1 (physical triplet X_k and auxiliary triplet Y_k)—into a unified geometric Landau-EFT. The Pontryagin protection, the spin-2 phase diagram, and the physical/auxiliary decomposition of the spin-1 sector were all integrated into a single framework. That paper establishes the baseline homogeneous mode dictionary for $S^3 \times S^1$ and the low-order EFT.

The role of paper III (LC) in the present work is precisely this: it provides the confirmed baseline mode dictionary and EFT for $S^3 \times S^1$, which the present paper retains while making the EC–Weyl corrections explicit and extending the comparison to all three topologies.

The objectives of the present paper are to extend this framework from the Levi–Civita (LC) connection to the Einstein–Cartan (EC) connection, and to carry out a systematic three-topology comparison. Specifically, we address:

1. What structure does the EC–Weyl coupling αC^2_{EC} have with respect to the torsion modes (AX/VT/MX)?

2. Is the Palatini protection (non-propagation of torsion) established in LC also preserved in EC?

3. How does the mode dictionary of the three topologies change in EC? In particular, does Nil^3 exhibit physics unique to EC?

4. How do the cross-term structures of squash/shear deformations and the EC–Weyl coupling reflect the geometric properties of each topology?

Answers to these four questions are given by the combination of exact symbolic computation with SymPy and analytic proofs, and are summarised as nine theorems.

1.2 Summary of Main Results

The main results of this paper are collected in nine theorems (Theorems 1–9). Here we describe the significance of each and their interrelations.

EC–Weyl coupling structure (§2): Theorem 1 provides the fundamental selection rule that the physical effect of the EC extension appears only in the MX background ($V\eta \neq 0$). This "AX/VT dropout" holds universally across all three topologies and clarifies the geometric mechanism of EC–Weyl interaction ($V \times \eta$ cross-terms activate the Weyl component). Theorem 3 complements this by establishing that no new cubic channel appears in the formal auxiliary MIXING sector, so the novel cubic structure of EC is localised in the physical η-V channel.

Protection structure of $S^3 \times S^1$ (§2, §3): Theorems 2 and 4 show that the mode dictionary of $S^3 \times S^1$ is essentially protected under the transition from LC to EC. Palatini protection (Theorem 2) preserves torsion non-propagation, and Palatini universality (Theorem 4) ensures that the spin-2/spin-1 mode spectrum is independent of the type of torsion background.

Triple triviality of $T^3 \times S^1$ (§4): The flatness of T^3 ($C^a_{bc} = 0$) makes all spin-2/spin-1 masses zero (§4.1), and algebraic constraints from AX/VT dropout and MX corrections coexist. Theorem 5 clarifies that the squash–shear cross-term seen in T^3 is a kinematic consequence of the topology-specific non-volume-preservation, distinct from the Weyl mass.

EC-specific physics of $\mathrm{Nil}^3 \times S^1$ (§5): Nil^3 is the only "non-conformally flat" manifold among the three topologies (non-zero background Weyl tensor), and this generates a positive EC-induced slice minimum on the $\eta = V = 0$ section for $\alpha < 0$ (Theorem 6). The low-energy EFT around this slice-minimum branch is organised through Theorem 7 (quintet splitting) and Theorem 8 (uniaxial mass splitting).

Three-topology comparison (§6): Theorem 9 classifies the geometric differences among the three topologies using the squash–shear cross-term as an indicator. The $\gamma = 1/2$ scaling theorem (§6.3) and the conformal-flatness theorem (§6.4) clarify the common geometric principle underlying the EC-specific physics of Nil^3.

1.3 LC→EC Update Summary

Table 1 summarises the changes from paper III (LC) to the present paper. "Preserved" indicates a structure established in LC that carries over to EC unchanged; "New" indicates a result first made explicit in the EC version.

Quantity / Structure	Paper III (LC)	This paper (EC)	Change
Palatini protection ($G_{\eta\eta} = G_{VV} = 0$)	✓	✓ (Thm. 2)	**Preserved**
AX/VT dropout	—	✓ all 3 topol. (Thm. 1)	**New**
Auxiliary C_δ	—	0 (all 3 topol., Thm. 3)	**New** (null result)
S^3 spin-2 mass (isotropic pt.)	$48\pi^2$ (conv. dep.)	$128\pi^2 Lr/(3\kappa^2) - 16384\pi^2 L\alpha/(3r)$ (Thm. 4)	**Preserved** (conv. unified)
S^3 spin-1 mass	Paper III §5	$16\pi^2 L^3/(\kappa^2 r)$ (Thm. 4)	**Preserved**
S^3 Pontryagin protection (AX, VT)	✓	✓ (α-independent)	**Preserved**
T^3 all masses	$= 0$ (flatness)	$= 0$ (Thm. 5 related)	**Preserved**
T^3 squash–shear cross-term	—	kinematic origin established (Thm. 5)	**New**
Nil^3 LC vacuum	absent (monotone)	absent (AX/VT: EC-LC diff. zero)	**Preserved**
Nil^3 EC slice minimum	—	$r_0 = (4\kappa/\sqrt{3})\sqrt{\lvert\alpha\rvert}$ on $\eta = V = 0$ (Thm. 6)	**New**
Nil^3 spin-2 splitting	—	$5 \to 0 + 2 + 1 + 1$ (Thm. 7)	**New**
Nil^3 spin-1 mass splitting	—	$m^2(\omega_2) \neq 0$ (Thm. 8)	**New**
ε-s cross-term classification	—	full 3-topology classification (Thm. 9)	**New**

Table 1: LC→EC update summary. "Preserved" means the LC structure carries over to EC; "New" means first established in the EC version.

1.4 Outline

The paper is organised as follows.

- Section 2 introduces the basic structure of the EC–Weyl coupling and establishes Theorems 1–3.
- Section 3 gives the mode dictionary for $S^3 \times S^1$ (Theorem 4).
- Section 4 establishes the mode dictionary for $T^3 \times S^1$ (Theorem 5).
- Section 5 develops the mode dictionary for $\mathrm{Nil}^3 \times S^1$ (Theorems 6–8).
- Section 6 performs the three-topology comparison (Theorem 9, $\gamma = 1/2$ scaling).
- Section 7 discusses the physical implications and future prospects.
- Section 8 is the conclusion.
- Appendix A collects notation and conventions; Appendix B gives the complete EFT coefficient list; Appendix C provides the detailed derivation of the ε-s cross-term; Appendix D gives the detailed proof of the auxiliary δ-sector null theorem.

2 EC–Weyl Coupling Structure

This section introduces the basic structure of the Einstein–Cartan connection and determines the dependence of the EC–Weyl scalar C^2_{EC} on the torsion modes. The main results are **Theorem 1 (AX/VT dropout)**, **Theorem 2 (Palatini protection, EC)**, and **Theorem 3 (δ-sector null theorem)**. The first two give the selection rule of the EC–Weyl coupling and the non-propagation of torsion; the third establishes that no new homogeneous cubic channel appears in the auxiliary MIXING sector.

2.1 EC Connection and Contortion

In Einstein–Cartan theory the connection $\omega^a{}_b$ is treated as a variable independent of the metric. Its difference from the Levi–Civita (LC) connection $\Gamma^a{}_{bc}$ is parametrised by the contortion $K^a{}_{bc}$:

$$\omega^a_{\mathrm{EC}\,bc} = \Gamma^a{}_{bc} + K^a{}_{bc}. \tag{1}$$

The contortion is determined from the antisymmetric torsion tensor $T^a{}_{bc} = 2\omega^a{}_{[bc]}$ by

$$K_{abc} = \frac{1}{2}\Big(T_{abc} + T_{bca} - T_{cab}\Big), \qquad K_{abc} = -K_{acb}. \tag{2}$$

For the homogeneous minisuperspace torsion ansatz we adopt three modes:

- **AX mode**: axial scalar η (traceless antisymmetric part of torsion), with $V = 0$ assumed.
- **VT mode**: vector-trace scalar V (trace part of torsion), with $\eta = 0$ assumed.
- **MX mode**: coexistence of AX and VT, i.e. $V \neq 0$ and $\eta \neq 0$.

The action is

$$S = \int \left[\frac{R_{\mathrm{EC}}}{2\kappa^2} + \theta \cdot \mathcal{L}_{\mathrm{NY}} + \alpha \cdot C^2_{\mathrm{EC}}\right] \sqrt{g}\, \mathrm{d}^4x, \tag{3}$$

where R_{EC} is the EC Ricci scalar, $\mathcal{L}_{\mathrm{NY}}$ is the Nieh–Yan density [5], and C^2_{EC} is the EC Weyl scalar [7]. The coupling constants are $\alpha \in \mathbb{R}$ (Weyl) and $\theta \in \mathbb{R}$ (NY). The effective potential $V_{\mathrm{eff}}(r, \eta, V; \alpha, \theta)$ is computed explicitly for each topology (script: `proofs/ax_vt_dropout_proof.py`).

2.2 Theorem 1: AX/VT Dropout

Theorem 1 (AX/VT dropout)**.** *In all of $S^3 \times S^1$, $T^3 \times S^1$, and $\mathrm{Nil}^3 \times S^1$, the following hold as exact symbolic identities verified by SymPy*:

1. ***AX mode*** *($V = 0$, $\eta \neq 0$): $C^2_{\mathrm{EC}} = C^2_{\mathrm{LC}}$ (no EC correction).*

2. ***VT mode*** *($\eta = 0$, $V \neq 0$): $C^2_{\mathrm{EC}} = C^2_{\mathrm{LC}}$ (no EC correction).*

In AX and VT backgrounds, the EC–Weyl coupling αC^2_{EC} coincides exactly with the LC case and no α-dependent new contribution appears.

Proof outline. Expanding $C^2_{\mathrm{EC}} - C^2_{\mathrm{LC}}$ symbolically for each topology and factoring, every EC correction term carries $V\eta$ as a common factor. Therefore in the AX mode ($V = 0$) the correction vanishes exactly, and in the VT mode ($\eta = 0$) it likewise vanishes (SymPy exact zero in all cases). □

Topology-specific forms. For $S^3 \times S^1$ and $T^3 \times S^1$:

$$C^2_{\mathrm{LC}}(S^3, T^3) = 0 \quad \text{(isotropic point, AX/VT background)}. \tag{4}$$

AX dropout then confirms $C^2_{\mathrm{EC}} = 0$.

For $\mathrm{Nil}^3 \times S^1$:

$$C^2_{\mathrm{LC}}(\mathrm{Nil}^3) = \frac{4}{3R^4} \neq 0 \quad \text{(non-conformal flatness)}, \tag{5}$$

and AX dropout gives $C^2_{\mathrm{EC}} = 4/(3R^4) = C^2_{\mathrm{LC}}$ (no EC correction).

T^3 squash+shear null test. For T^3, $C^2_{\mathrm{EC}} = 0$ holds for all ε, s because $C^a_{bc} = 0$ identically. These six cases (3 topologies × 2 backgrounds) are all confirmed as exact zero by SymPy (scripts: `proofs/ax_vt_dropout_proof.py`, `proofs/t3_flatness_null_test.py`).

2.3 MX Mode: EC-Specific Weyl Coupling

As the converse of Theorem 1, in the MX background ($V \neq 0, \eta \neq 0$) an EC-specific Weyl coupling appears.

For $S^3 \times S^1$ and $T^3 \times S^1$:

$$C^2_{\mathrm{EC}} = C^2_{\mathrm{LC}} + \frac{16V^2\eta^2}{3r^2}. \tag{6}$$

For $\mathrm{Nil}^3 \times S^1$:

$$C^2_{\mathrm{EC}} = C^2_{\mathrm{LC}} + \frac{16V^2\eta^2}{3R^2} = \frac{4}{3R^4} + \frac{16V^2\eta^2}{3R^2}. \tag{7}$$

In both cases the additional term is proportional to $(V\eta)^2$, so the torsion mixing product modifies the effective potential via the EC–Weyl coupling:

$$\Delta V^{\mathrm{EC}}_{\mathrm{eff}} = \alpha \cdot \frac{16V^2\eta^2}{3r^2} \cdot \mathrm{Vol}(M^3 \times S^1). \tag{8}$$

This EC-specific term is the source of effective-potential deformation in the MX background. Note, however, that the Nil^3 EC slice minimum discussed in §5 originates from $C^2_{\mathrm{LC}}(\mathrm{Nil}^3) \neq 0$ on the $\eta = V = 0$ section, which is independent of this MX term.

Physical interpretation. AX/VT dropout means that torsion alone does not generate new contributions to the Weyl tensor. The essence of the EC–Weyl interaction lies in the "mixing product" $V \times \eta$ of torsion, understood geometrically as the mechanism by which an asymmetric combination of contortion activates Weyl components. This selection rule is universal across all three topologies and follows from the general algebraic structure of the EC connection.

2.4 Theorem 2: Palatini Protection (EC Version)

Theorem 2 (Palatini protection, EC)**.** *Under the Einstein–Cartan connection,*

$$G^{\mathrm{EC}}_{\eta\eta} = G^{\mathrm{EC}}_{VV} = 0 \tag{9}$$

holds exactly, where $G^{\mathrm{EC}}_{AB} = \partial^2 \mathcal{L}_{\mathrm{EC}} / \partial(\partial_z \Phi^A) \partial(\partial_z \Phi^B)$ *is the field-space metric (kinetic metric) of the minisuperspace.*

Proof outline (velocity method). The EC Weyl scalar in the MX background, $C^2_{\mathrm{EC}}(S^3, T^3, \mathrm{MX}) = 16V^2\eta^2/(3r^2)$, is an algebraic function of (r, η, V) and contains no velocity fields $\partial_z \eta$, $\partial_z V$. Therefore, computing $G_{\eta\eta}$, G_{VV} by the velocity method yields only terms that vanish in the $\Delta z \to 0$ limit, and no kinetic term for torsion is generated. Full symbolic verification for each topology is given in script `proofs/palatini_protection_proof.py`. □

Topology	G_{rr} (LC)	G_{rr} (EC)	Difference
S^3	59.22	59.22	0
T^3	4675.6	4675.6	≈ 0
Nil^3	5195.2	5195.2	≈ 0

Table 2: Kinetic metric G_{rr} for the radial mode: LC vs. EC. The graviton kinetic structure is unchanged under the EC extension.

Invariance of r-stiffness. The kinetic structure of the spin-2 graviton is also invariant under EC.

2.5 Theorem 3: δ-Sector Null Theorem

In the EC extension, the new nonlinear coupling

$$\frac{\partial^3 V_{\rm EC}}{\partial\alpha\,\partial\eta\,\partial V} \neq 0 \tag{10}$$

appears in the physical (η, V) channel (§3.4). A separate question is whether an independent cubic channel also arises in the auxiliary MIXING variable δ_k. We define

$$C_\delta(M^3; \text{mode}) := \left.\frac{\partial^3 V_{\rm eff}}{\partial\delta_0\,\partial\delta_1\,\partial\delta_2}\right|_{\delta_0=\delta_1=\delta_2=0}, \tag{11}$$

where $M^3 = S^3, T^3, \text{Nil}^3$ and mode = AX/VT/MX.

Theorem 3 (δ-sector null theorem)**.** *For all three topologies and all torsion backgrounds (AX, VT, MX),*

$$C_\delta(M^3; \mathit{mode}) = 0 \qquad \mathit{and} \qquad \frac{\partial C_\delta}{\partial\alpha} = 0 \tag{12}$$

hold as exact SymPy results. The EC extension generates no new homogeneous cubic channel in the auxiliary δ-sector.

Proof outline. With `fiber_mode = MIXING` activated in the unified engine, we construct $V_{\rm eff}$ for each topology and torsion background, then compute the third derivative $\partial_{\delta_0}\partial_{\delta_1}\partial_{\delta_2} V_{\rm eff}|_{\delta=0}$ by direct symbolic differentiation. All nine cases give exact zero coefficients, from which $\partial_\alpha C_\delta = 0$ follows automatically. The key point is not that EC has no new effects, but that the new cubic structure activated by EC is localised in the physical η-V channel rather than the auxiliary δ-sector. □

Topology-wise consequences.

- S^3: The formal MIXING sector is nontrivially available, yet $C_\delta(S^3) = 0$. Hence the α-cubic of §3.4 appears only in the η-V sector.
- T^3: Flatness trivialises most of the mode dictionary. Even the formal MIXING test gives $C_\delta(T^3) = 0$.
- Nil^3: Despite the non-conformal flatness and the EC false vacuum, $C_\delta(\text{Nil}^3) = 0$. The EC-specific physics of Nil^3 resides in the background Weyl and the (r, η, V) sector.

Detailed derivations and the exact check of all nine cases are given in Appendix D and script `proofs/c_delta_null_proof.py`.

3 Mode Dictionary: $S^3 \times S^1$ under EC–Weyl

This section determines the effect of the EC–Weyl coupling on the homogeneous geometric degrees of freedom of $S^3 \times S^1$. Since the non-propagation of torsion in the spin-0 sector is already established by Theorem 2, we focus here on the specific consequences for S^3: Theorem 4 (Palatini universality) and its reflection in the mode dictionary.

3.1 Theorem 4: Palatini Universality

Theorem 4 (Palatini universality)**.** *In $S^3 \times S^1$, the spin-2 and spin-1 masses in the AX and VT backgrounds are identical (SymPy exact)*:

***Spin-2 mass (five-fold degenerate)*:**

$$\mu^2_{\text{spin2}}(S^3) = \frac{128\pi^2 Lr}{3\kappa^2} - \frac{16384\pi^2 L\alpha}{3r}. \tag{13}$$

***Spin-1 mass (three-fold degenerate)*:**

$$m^2_{\omega_k}(S^3) = \frac{16\pi^2 L^3}{\kappa^2 r}, \qquad k = 0, 1, 2. \tag{14}$$

Both expressions are independent of η and V: the mode spectrum does not depend on the type of torsion background.

The VT−AX difference for spin-2 mass is exactly zero (SymPy exact). The LC contribution ($\alpha = 0$) to the spin-2 mass is $128\pi^2 Lr/(3\kappa^2)$ and the EC correction is $-16384\pi^2 L\alpha/(3r)$, the latter depending only on the topology (S^3 geometry) and not on η or V.

Remark 1. The numerical evaluation at $r = 3$, $L = \kappa = 1$, $\alpha = 0$ gives $\mu^2_{\text{spin2}} = 128\pi^2 \approx 1263.3$ and $m^2_{\omega_k} = 16\pi^2/3 \approx 52.6$. The LC paper III uses $r = 1$-based notation, giving $128\pi^2/3 \approx 421.2$ for the spin-2 mass. The discrepancy with the value $48\pi^2$ quoted there arises from the choice of radius combined with conventions for the spin-2 basis/Hessian.

Relation to the physical spin-1 mode. Here $m^2_{\omega_k} = 16\pi^2 L^3/(\kappa^2 r)$ is the effective-potential Hessian along the twist direction ω_k ($\partial^2 V_{\text{eff}}/\partial\omega_k^2|_0$), a kinematic quantity used as the indicator for Palatini universality (AX=VT) and for the three-topology comparison. It is *not* the full physical spin-1 eigenvalue: in the full spin-1 system, where the twist ω and the mixing δ couple and the field-space metric is rank-deficient, the physical mode obtained after Schur-eliminating the auxiliary has a *negative* effective mass squared (tachyonic, non-ghost; Paper III §5). This does not affect Theorem 2 or the topology comparison, which use only the kinematic twist Hessian.

Physical interpretation. Palatini universality means that the difference between the AX background ($\eta \neq 0$) and the VT background ($V \neq 0$) has no effect on the spin-2/spin-1 mass spectrum. The kinematic part coming from the S^3 geometry ($C^i_{jk} = 4\varepsilon_{ijk}/r$) determines the masses, while torsion deforms only the shape of the potential (a consequence of Palatini protection, Theorem 2). Script: `paper03ec/s3_vt_spin_masses.py`.

3.2 Mode Dictionary Table ($S^3 \times S^1$)

Table 3 summarises the mode dictionary of $S^3 \times S^1$ under EC–Weyl.

Quantity	AX ($V = 0$, $\eta \neq 0$)	VT ($\eta = 0$, $V \neq 0$)	MX ($V \neq 0$, $\eta \neq 0$)
C^2_{EC}	$= C^2_{\mathrm{LC}} = 0$	$= C^2_{\mathrm{LC}} = 0$	$16V^2\eta^2/(3r^2)$
AX/VT dropout	✓	✓	×
spin-0: r	propagating	propagating	propagating
spin-0: η, V	η non-prop., $V = 0$ (Thm. 2)	$\eta = 0$, V non-prop. (Thm. 2)	both non-prop. (Thm. 2)
spin-2 mass μ^2	$128\pi^2 Lr/(3\kappa^2) - 16384\pi^2 L\alpha/(3r)$	same as AX (Thm. 4)	—
spin-1 mass m^2_ω	$16\pi^2 L^3/(\kappa^2 r)$ (3-fold)	same as AX (Thm. 4)	—

Table 3: $S^3 \times S^1$ mode dictionary under EC–Weyl.

Regarding the MX sector of $S^3 \times S^1$: the supplementary script `paper03ec/s3_mx_vacuum_search.py` checked the stationarity conditions and Hessian within the range $|\eta| \leq 3$. No stable full stationary vacuum satisfying both $\nabla_{(r,\eta,V)} V_{\mathrm{eff}} = 0$ and positive-definite Hessian was found. Real stationary points may appear on the $\alpha > 0$ side, but all of them have at least one negative eigenvalue and are saddle points. The finite-radius well appearing in a fixed-V slice is a one-dimensional minimum in the r direction and does not constitute a full stationary vacuum.

3.3 Pontryagin Structure: AX = Plücker, VT = Pfaffian, MX $\neq 0$

The Pontryagin protection structure is inherited from paper III.

AX mode (Plücker protection): When $V = 0$, the Pontryagin density $P = \langle R, \star R \rangle$ vanishes identically by a Plücker relation. AX torsion contributes no antisymmetric part to the Riemann tensor that could activate P.

VT mode (Pfaffian protection): When $\eta = 0$, P vanishes via a Pfaffian-type identity. VT torsion modifies only the symmetric part of the Riemann tensor, leaving the antisymmetric combination that enters P zero.

MX mode: When $V \neq 0, \eta \neq 0$, the Pontryagin density at the isotropic point is (from paper III):

$$P_0 = \frac{2V\eta(V^2r^2 + 9\eta^2 - 36)}{9r^3}. \tag{15}$$

Effect of EC. The Weyl coupling α enters only the action, not the definition of the Pontryagin density P itself. Therefore the AX/VT/MX Pontryagin classification is unchanged from paper III, and what changes in the EC extension is the effective potential V_{eff} side.

3.4 EC-Specific Modification under EC–Weyl

Preservation of AX/VT stability. Under EC correction $\alpha < 0$, the structure of the AX/VT stable vacua is unchanged from LC. By the AX/VT dropout (Theorem 1), $C^2_{\text{EC}} = C^2_{\text{LC}}$, so the Weyl coupling does not modify the effective potential.

EC-specific cubic coupling (α-cubic). As a new nonlinear coupling absent in LC, the following appears:

$$\frac{\partial^3 V_{\text{EC}}}{\partial\alpha\,\partial\eta\,\partial V} = -\frac{128\pi^2 V\eta r}{3} \neq 0 \qquad (S^3). \tag{16}$$

This gives the EC-specific cubic coefficient in the homogeneous EFT with respect to α, a cubic channel absent in LC (script: `paper03ec/ec_cubic_vertex.py`). This term does not appear in the second-order mode dictionary of Table 3 and enters only at cubic order in the homogeneous EFT. The coefficient corresponds to the $S^3 \times S^1$ volume convention $\text{Vol}(S^3 \times S^1) = 2\pi^2 L r^3$ of Appendix A. For comparison, the NY-cubic channel $\partial^3 V/\partial\theta\partial\eta\partial V = 4\pi^2 r^2$ is invariant under EC (a consequence of Palatini protection).

In the auxiliary formal MIXING sector, on the other hand,

$$C_\delta(S^3) = 0, \qquad \frac{\partial C_\delta(S^3)}{\partial\alpha} = 0 \tag{17}$$

(Theorem 3, Appendix D). Therefore the new homogeneous cubic structure activated by EC extension is confined to the physical η-V channel, and the δ-sector remains null.

4 Mode Dictionary: $T^3 \times S^1$ under EC–Weyl

This section analyses the effect of the EC–Weyl coupling on the homogeneous geometric degrees of freedom of $T^3 \times S^1$. The flatness of T^3 ($C^a_{bc} = 0$) causes a triple triviality, and we establish Theorem 5 (torsion–volume coupling) as the origin of the squash–shear cross-term.

4.1 T^3 Flatness Theorem

$T^3 = \mathbb{R}^3/\mathbb{Z}^3$ is a flat torus with

$$C^a_{bc}(T^3) = 0 \quad \forall\, a, b, c. \tag{18}$$

This flatness causes the Weyl tensor to vanish, setting all spin-2/spin-1 masses to zero.

Spin-2 mass (squash direction). By exact SymPy computation,

$$\left.\frac{\partial^2 V}{\partial \varepsilon^2}\right|_{\varepsilon=0} = 0 \quad \forall\, \eta, V, \alpha. \tag{19}$$

With $C^a_{bc} = 0$, the Weyl tensor vanishes and no geometric mass term for spin-2 arises. Script: `paper03ec/t3_mode_dictionary.py`.

Spin-2 mass (shear direction). Similarly,

$$\left.\frac{\partial^2 V}{\partial s^2}\right|_{s=0} = 0 \quad \forall\, \eta, V, \alpha \tag{20}$$

(SymPy exact zero; script: `proofs/t3_flatness_null_test.py`).

Spin-1 mass. With $C^a_{bc} = 0$, all spin-1 components are massless: $m^2_\omega = 0$.

4.2 Mode Dictionary Table ($T^3 \times S^1$)

Quantity	AX ($V = 0$, $\eta \neq 0$)	VT ($\eta = 0$, $V \neq 0$)	MX ($V \neq 0$, $\eta \neq 0$)
C^2_{EC}	$= C^2_{\mathrm{LC}} = 0$	$= C^2_{\mathrm{LC}} = 0$	$16V^2\eta^2/(3r^2)$
AX/VT dropout	✓	✓	×
spin-0: r	propagating	propagating	propagating
spin-0: η, V	η non-prop., $V = 0$	$\eta = 0$, V non-prop.	both non-prop.
spin-2 mass μ^2	0 (T^3 flatness)	0 (T^3 flatness)	0 (T^3 flatness)
spin-1 mass m^2_ω	0 (massless)	0	0

Table 4: $T^3 \times S^1$ mode dictionary under EC–Weyl.

4.3 Triple Triviality of T^3

The effect of the EC–Weyl coupling in $T^3 \times S^1$ is trivial in three senses.

(i) AX dropout. From $C^a_{bc} = 0$ we get $C^2_{\mathrm{LC}} = 0$; Theorem 1 (AX dropout) then gives $C^2_{\mathrm{EC}} = C^2_{\mathrm{LC}} = 0$. The coupling α does not modify the effective potential in the AX background.

(ii) VT dropout. Similarly, $\eta = 0$ gives Theorem 1 (VT dropout) and $C^2_{\mathrm{EC}} = 0$. In the VT background too, α is inactive.

(iii) MX correction. In the MX background ($V\eta \neq 0$) where AX/VT dropout does not apply, $C^2_{\mathrm{EC}} = 16V^2\eta^2/(3r^2) \neq 0$. For $\alpha < 0$ this correction gives $\Delta V^{\mathrm{EC}}_{\mathrm{eff}} > 0$, pushing the potential upward. Whether a finite MX critical point exists is examined algebraically in §4.5.

The auxiliary EFT coefficient is also

$$C_\delta(T^3) = 0, \qquad \frac{\partial C_\delta(T^3)}{\partial \alpha} = 0 \tag{21}$$

(Theorem 3, Appendix D). Hence in T^3, neither the physical sector nor the auxiliary sector develops a new homogeneous cubic channel from the EC extension.

4.4 Theorem 5: Torsion–Volume Coupling

Theorem 5 (Torsion–volume coupling)**.** *In $T^3 \times S^1$, the mixed second derivative of the squash and shear parameters satisfies*

$$\left.\frac{\partial^2 V_{\mathrm{eff}}}{\partial\varepsilon\,\partial s}\right|_{\varepsilon=s=0} = \frac{48\pi^4 L\eta^2 r}{\kappa^2} \neq 0 \quad (\eta \neq 0). \tag{22}$$

The origin of this cross-term is the kinematic mixing of the volume element $(1+\varepsilon)(1+s)$, independent of the Weyl mass.

Proof outline. The effective potential of T^3 takes the form (AX background, $V = 0$):

$$V^{T^3}_{\mathrm{eff}}(\varepsilon, s) \supset \frac{16\pi^4 Lr}{\kappa^2} \cdot 9\eta^2 \cdot (1+\varepsilon)(1+s) + \cdots \tag{23}$$

(with $C^a_{bc} = 0$ the Weyl contribution is absent, and the torsion kinetic term $\propto \eta^2$ is multiplied by the volume factor $(1+\varepsilon)(1+s)$). The mixed derivative yields:

$$\left.\frac{\partial^2 V^{T^3}_{\mathrm{eff}}}{\partial\varepsilon\,\partial s}\right|_{\varepsilon=s=0} = \frac{16\pi^4 Lr}{\kappa^2} \cdot 9\eta^2 \cdot 1 = \frac{48\pi^4 L\eta^2 r}{\kappa^2}. \tag{24}$$

Confirmation that this is not a Weyl mass. In the AX background ($V = 0$), $C^a_{bc} = 0$ gives $C^2_{\mathrm{LC}} = 0$, and Theorem 1 (AX dropout) gives $C^2_{\mathrm{EC}} = 0$, so this cross-term is α-independent and carries no Weyl-mass contribution. Numerical check (AX background): the cross-term value is identical for $\alpha = 0$ and $\alpha = -1$ (SymPy exact). In the MX background ($V \neq 0$), α-dependent terms join the cross-term via $C^2_{\mathrm{EC}} = 16V^2\eta^2/(3r^2) \neq 0$, but the spin-2 null test ($\partial^2 V/\partial\varepsilon^2 = 0$, $\partial^2 V/\partial s^2 = 0$) continues to hold (because C^2_{EC} is independent of ε, s). Scripts: `proofs/t3_flatness_null_test.py`, `paper03ec/squash_shear_cross_term.py`. □

4.5 Absence of MX Critical Points

We examine critical points of the T^3 MX sector for $\alpha < 0$. Solving $\partial V_{\mathrm{eff}}/\partial\eta = \partial V_{\mathrm{eff}}/\partial V = 0$ symbolically by SymPy, the MX solution is

$$V^* = \pm\frac{3}{4}\sqrt{\frac{1}{\alpha}}, \qquad \eta^* = \pm\frac{r}{4}\sqrt{\frac{1}{\alpha}}, \tag{25}$$

which is purely imaginary for $\alpha < 0$ (no real solution). Therefore **no finite real MX critical point exists in T^3 for $\alpha < 0$.** This is consistent with the analysis in §4.3 that the MX correction pushes the potential upward, giving no new homogeneous local minimum in the MX sector of T^3. Script: `paper03ec/t3_mx_vacuum_search.py`.

5 Mode Dictionary: $\mathrm{Nil}^3 \times S^1$ under EC–Weyl

This section analyses in detail the effect of the EC–Weyl coupling on the homogeneous geometric degrees of freedom of $\mathrm{Nil}^3 \times S^1$. Among the three topologies, Nil^3 is the only non-conformally flat manifold, and this property generates—via the EC–Weyl coupling—an EC slice minimum (Theorem 6), a spin-2 quintet splitting (Theorem 7), and a uniaxial mass splitting (Theorem 8), all absent in LC.

5.1 Geometric Properties of Nil^3

The geometry of Nil^3 (the three-dimensional Heisenberg manifold) is characterised by a single non-zero structure constant:

$$C^2_{01}(\mathrm{Nil}^3) = \frac{1}{R}, \tag{26}$$

where R is the scale parameter of Nil^3. We adopt the convention $[E_0, E_1] = (1/R)E_2$. In the implementation, the Maurer–Cartan form $\mathrm{d}\sigma^2 = -\sigma^0 \wedge \sigma^1$ is used as the basis, so the frame coefficient is $C[2,0,1] = -(1+\varepsilon)^{-4/3}(1+s)^{-2}/R$. Both expressions describe the same Heisenberg structure and differ only by a convention choice. This single-component Lie structure is the source of the spin-2 quintet splitting and uniaxial mass splitting discussed below.

Non-conformal flatness. The background ($\eta = V = 0$) Weyl scalar is

$$C^2_{\mathrm{LC}}(\mathrm{Nil}^3,\, \eta = V = 0) = \frac{4}{3R^4} \neq 0. \tag{27}$$

This non-conformal flatness activates the EC–Weyl coupling $\alpha \cdot C^2_{\mathrm{EC}}$ even on the $\eta = V = 0$ section, generating the EC slice minimum (§5.2).

The auxiliary cubic coefficient satisfies

$$C_\delta(\mathrm{Nil}^3) = 0, \qquad \frac{\partial C_\delta(\mathrm{Nil}^3)}{\partial \alpha} = 0 \tag{28}$$

(Theorem 3, Appendix D). The EC-specific physics of Nil^3 therefore appears in the background Weyl and the (r, η, V) sector, not as a new δ-sector cubic channel.

5.2 Theorem 6: Nil^3 EC Slice Minimum

Theorem 6 (Nil^3 EC slice minimum)**.** *For $\alpha < 0$, the homogeneous effective potential of $\mathrm{Nil}^3 \times S^1$ restricted to the $\eta = V = 0$ section supports a positive EC-induced slice*

minimum:

$$r_0 = \frac{4\kappa}{\sqrt{3}}\sqrt{|\alpha|}, \qquad V_0 = \frac{32\sqrt{3}\,\pi^4 L a}{3\kappa} > 0, \quad (\alpha = -a^2,\ a > 0). \tag{29}$$

Moreover, the stationary point $(r, \eta, V) = (r_0, 0, 0)$ *is a local minimum of the full homogeneous potential for* $|\kappa^2\theta_{\rm NY}| < 1$*, marginal for* $|\kappa^2\theta_{\rm NY}| = 1$*, and a saddle for* $|\kappa^2\theta_{\rm NY}| > 1$*.*

Terminological note. Earlier drafts referred to this point as a "false vacuum." In the present revision we use "EC slice minimum" to emphasise that it is first identified on the $\eta = V = 0$ section and that full homogeneous stability requires the explicit Hessian criterion stated above.

Proof. By AX/VT dropout (Theorem 1), in the AX background ($V = 0$) and the VT background ($\eta = 0$), $C^2_{\rm EC} = C^2_{\rm LC}$ holds, so the EC extension does not shift the LC stationarity conditions. (Since $C^2_{\rm LC}(\mathrm{Nil}^3) \neq 0$, the α-dependence through the LC–Weyl term remains, but this is already present in LC.)

On the $\eta = V = 0$ isotropic section, SymPy confirms the two-term structure:

$$V_{\rm eff}^{Nil^3}(r,\, \eta = 0,\, V = 0;\, \alpha) = \frac{4\pi^4 L r}{\kappa^2} - \frac{64\pi^4 L\alpha}{3r}. \tag{30}$$

The first term is the LC contribution (power $n = 1$) and the second is the EC–Weyl contribution (power $m = -1$).

The stationarity condition $\partial V_{\rm eff}/\partial r = 0$ gives

$$\frac{4\pi^4 L}{\kappa^2} + \frac{64\pi^4 L\alpha}{3r^2} = 0 \quad \Rightarrow \quad r_0^2 = -\frac{16\alpha\kappa^2}{3} = \frac{16|\alpha|\kappa^2}{3}, \quad r_0 = \frac{4\kappa}{\sqrt{3}}\sqrt{|\alpha|}. \tag{31}$$

The vacuum energy is

$$V_0 = V_{\rm eff}(r_0) = \frac{32\sqrt{3}\,\pi^4 L a}{3\kappa} > 0 \quad (\alpha = -a^2 < 0), \tag{32}$$

establishing a positive slice minimum.

Full homogeneous stability is decided by the Hessian at the stationary point:

$$H = \left.\frac{\partial^2 V_{\rm eff}}{\partial(r,\eta,V)^2}\right|_{(r_0,0,0)} = \begin{pmatrix} \dfrac{2\sqrt{3}\pi^4 L}{\kappa^3\sqrt{-\alpha}} & 0 & 0 \\ 0 & \dfrac{128\sqrt{3}\pi^4 L\sqrt{-\alpha}}{\kappa} & -\dfrac{512\pi^4 L\alpha\kappa^2\theta_{\rm NY}}{3} \\ 0 & -\dfrac{512\pi^4 L\alpha\kappa^2\theta_{\rm NY}}{3} & \dfrac{2048\sqrt{3}\pi^4 L\kappa(-\alpha)^{3/2}}{27} \end{pmatrix}. \tag{33}$$

The r-direction principal minor is always positive, while the remaining (η, V) block has

determinant

$$\det H_{(\eta,V)} = \frac{262144\pi^8 L^2\alpha^2}{9}\left(1-\kappa^4\theta_{\mathrm{NY}}^2\right). \tag{34}$$

Sylvester's criterion therefore gives the stated condition $|\kappa^2\theta_{\mathrm{NY}}| < 1$ for a full homogeneous local minimum, with the equality case marginal and the larger-$|\theta_{\mathrm{NY}}|$ regime a saddle. Script: `proofs/nil3_slice_minimum_stability.py`. □

α	r_0 (analytic)	r_0 (numeric)	Error	V_0 (analytic)	V_0 (numeric)	Error
-0.05	0.516398	0.516400	0.0004%	402.415	402.415	$< 10^{-6}$%
-0.50	1.632993	1.633000	0.0004%	1272.547	1272.547	$< 10^{-6}$%
-1.00	2.309401	2.309400	0.0000%	1799.653	1799.653	$< 10^{-6}$%
-3.00	4.000000	4.000000	0.0000%	3117.091	3117.091	0%

Table 5: Analytic vs. numeric comparison for the Nil^3 EC slice minimum (with $\kappa = L = 1$).

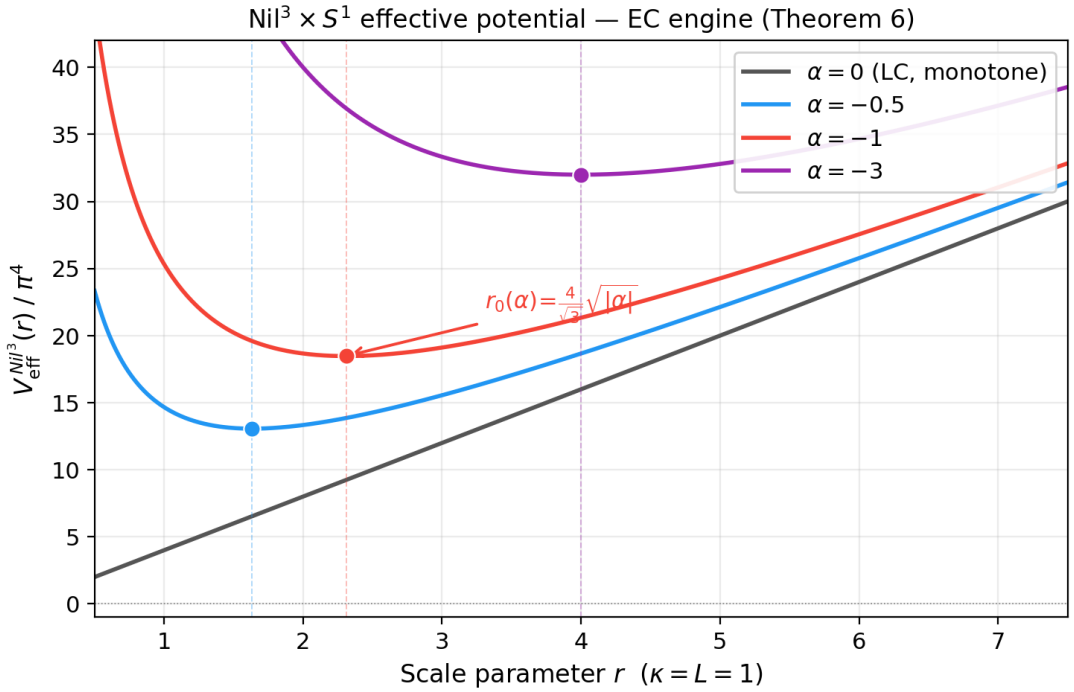


Figure 1: Nil^3 effective potential $V_{\mathrm{eff}}(r)$ on the $\eta = V = 0$ section for $\alpha = -1$, $\kappa = L = 1$. The LC part ($\alpha = 0$, dashed) is monotonically increasing; the EC–Weyl term ($-64\pi^4 L\alpha/(3r)$) creates a slice minimum at $r_0 = 2.309$ (solid).

Numerical verification.

5.3 Theorem 7: Nil^3 Spin-2 Quintet Splitting

Theorem 7 (Nil^3 quintet splitting)**.** *The spin-2 quintet (5 components) around the* $\mathrm{Nil}^3 \times S^1$ *EC slice-minimum branch splits as*

$$5 \;\to\; (1)_0 + (2)_{q_4=q_5} + (1)_\varepsilon + (1)_s, \tag{35}$$

where the subscript on each bracket denotes the multiplicity.

Complete splitting pattern. At $a = \kappa = L = 1$, $r_0 = 4/\sqrt{3}$, $\alpha = -1$:

Component	DOF	Mass m^2	Physical interpretation
q_3	off-diagonal (0–1)	**0** (zero mode)	Geometric protection by Nil^3 symmetry
q_4	off-diagonal (0–2)	1.080×10^4	Degenerate mass
q_5	off-diagonal (1–2)	1.080×10^4	Degenerate with q_4 (Nil^3 symmetry)
ε	squash	3.919×10^4	Axial deformation (U(1) breaking)
s	shear	8.278×10^4	Shear deformation

Table 6: Nil^3 spin-2 quintet splitting at the EC slice minimum ($a = \kappa = L = 1$, $r_0 = 4/\sqrt{3}$).

The squash–shear block eigenvalues are $\lambda_+ = 1.137 \times 10^5$ and $\lambda_- = 8.280 \times 10^3$ (SymPy confirmed).

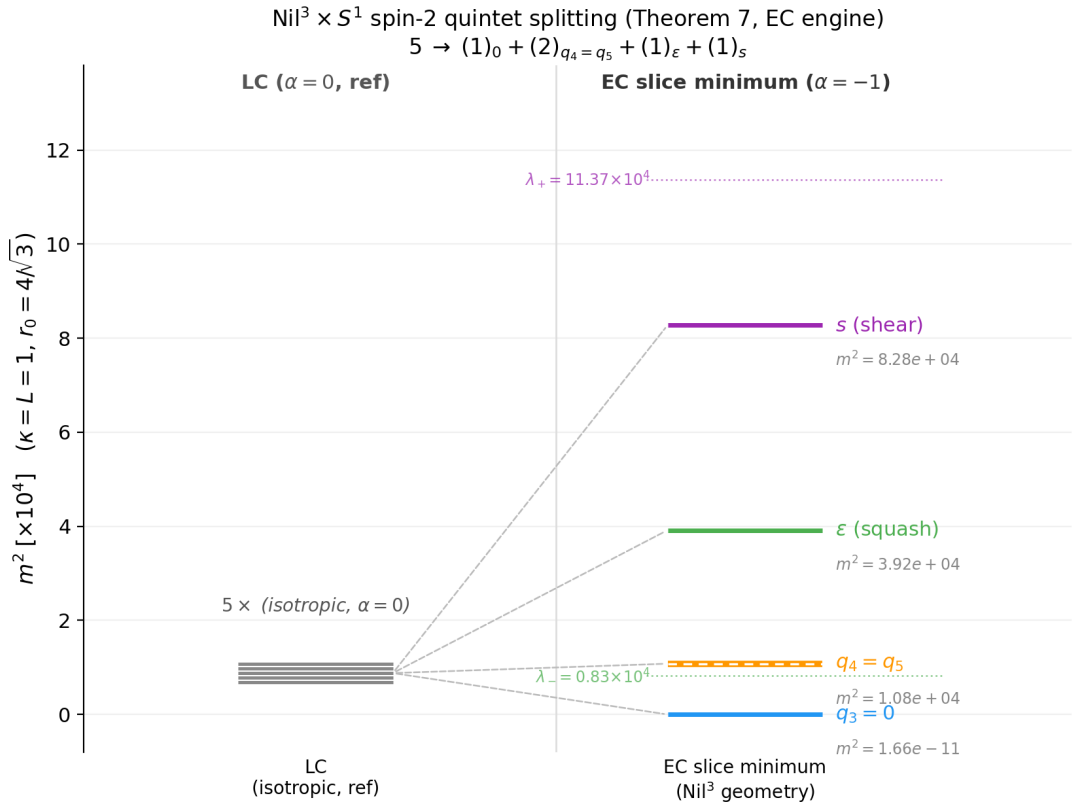


Figure 2: Nil^3 spin-2 quintet splitting $5 \to 0 + 2 + 1 + 1$ at the EC slice minimum. The zero mode q_3 is geometrically protected by the one-dimensional Lie structure of Nil^3, and the q_4–q_5 degeneracy reflects the symmetric role of e^0 and e^1 with respect to C^2_{01}.

Geometric interpretation of $q_3 = 0$. The G_3 transformation is a hyperbolic rotation in the (e^0, e^1) plane:

$$e^0 \to c_3 e^0 + s_3 e^1, \quad e^1 \to s_3 e^0 + c_3 e^1 \quad (c_3^2 - s_3^2 = 1). \tag{36}$$

Under this transformation, the sole structure constant C^2_{01} transforms as

$$C^2_{01} \to \frac{c_3^2 - s_3^2}{R} = \frac{1}{R} \tag{37}$$

(hyperbolic identity), and is therefore invariant. Consequently V_{eff} has no q_3 dependence and $m^2(q_3) = 0$. This is a geometric symmetry protection by the one-dimensional Lie structure of Nil^3 (C^2_{01} only), not a tachyonic instability.

Interpretation of $q_4 = q_5$ degeneracy. G_4 and G_5 are hyperbolic rotations in the (0–2) and (1–2) planes, respectively. In Nil^3, e^2 is the "result axis" of C^2_{01}, and e^0, e^1 play symmetric roles with respect to C^2_{01}. This symmetry forces $m^2(q_4) = m^2(q_5)$.

Script: `paper03ec/nil3_spin2_quintet_splitting.py`.

5.4 Theorem 8: Uniaxial Mass Splitting

Theorem 8 (Uniaxial mass splitting)**.** *In the spin-1 sector of* $\mathrm{Nil}^3 \times S^1$*, only one axis acquires a mass via the directional selectivity of* C_{01}^2:

$$m^2(\omega_2) = \frac{24\pi^4 L^3 R^2 - 256\pi^4 L^3 \alpha\kappa^2}{3R^3\kappa^2} \neq 0, \qquad m^2(\omega_0) = m^2(\omega_1) = 0. \tag{38}$$

At the EC slice minimum $r_0 = 4\kappa a/\sqrt{3}$ *(*$\alpha = -a^2$*):*

$$m^2(\omega_2)\Big|_{r_0} = \frac{6\sqrt{3}\,\pi^4 L^3}{a\kappa^3}. \tag{39}$$

Physical interpretation. C_{01}^2 corresponds to the non-commutativity in the (0–1) plane direction. ω_2 is the twist in the (0–1) direction, and only this component acquires a mass from $C_{01}^2 \neq 0$. ω_0 and ω_1 are commutative directions (with respect to C_{01}^2) and remain massless. This structure is termed uniaxial mass splitting. The one-dimensional Lie algebra structure of Nil^3 (C_{01}^2 only non-zero) gives mass to the unique ω_2 among the spin-1 triplet, while keeping the other two massless.

5.5 EFT around the Nil^3 EC Slice Minimum

We determine the quadratic effective action around the EC slice minimum $r_0 = 4\kappa a/\sqrt{3}$. The quoted EFT below uses the standard benchmark $\theta_{\mathrm{NY}} = 0$, for which the spin-0 Hessian returns to a diagonal block.

DOF	Quadratic coefficient (analytic)	Numeric ($a = \kappa = L = 1$)	α-dependence
r (radial)	$2\sqrt{3}\pi^4 L/(a\kappa^3)$	337.4	$\propto 1/a$ (light)
ω_2 (non-comm. spin-1)	$6\sqrt{3}\pi^4 L^3/(a\kappa^3)$	1012	$\propto 1/a$ (light)
η (axial torsion; non-prop.)	$128\sqrt{3}\pi^4 La/\kappa$	21,596	$\propto a$ (heavy)
spin-2 squash ε	$6272\sqrt{3}\pi^4 La/(27\kappa)$	39,192	$\propto a$ (heavy)
ω_0, ω_1	0	0	massless
q_3 (zero mode)	0	0	zero mode
q_4, q_5	$m^2(q_4) = m^2(q_5)$	10,798	—

Table 7: Quadratic coefficient spectrum of the Nil^3 EC slice-minimum EFT at $a = \kappa = L = 1$. The η row gives the potential curvature in the η direction under Palatini protection ($G_{\eta\eta} = 0$), not a propagating mass.

Full quadratic coefficient spectrum (numerical values at $a = \kappa = L = 1$**).**

Two-group hierarchy. "Geometric modes" (r, ω_2) scale as $\propto 1/a$ (lighter for large $|\alpha|$), while "torsion/curvature modes" (η-direction curvature, spin-2) scale as $\propto a$ (heavier for large $|\alpha|$).

The quadratic effective action is

$$V_{\text{eff}}^{\text{Nil}^3,\text{EFT}}(\delta\Phi) \approx V_0 + \frac{1}{2}\Bigg[\frac{2\sqrt{3}\pi^4 L}{a\kappa^3}(\delta r)^2 + \frac{128\sqrt{3}\pi^4 La}{\kappa}(\delta\eta)^2 + \frac{6272\sqrt{3}\pi^4 La}{27\kappa}(\delta\varepsilon)^2 + \frac{6\sqrt{3}\pi^4 L^3}{a\kappa^3}(\delta\omega_2)^2\Bigg] + O(\delta^3) \tag{40}$$

with the cross-term $\partial^2 V/\partial r\partial\eta|_{r_0,0} = 0$. Scripts: `paper03ec/nil3_false_vacuum_eft.py`.

5.6 Mode Dictionary Table ($\text{Nil}^3 \times S^1$)

Quantity	AX ($V=0$, $\eta\neq 0$)	VT ($\eta=0$, $V\neq 0$)	EC slice min. ($\eta=V=0$, $\alpha<0$)
C^2_{EC}	$4/(3R^4)$	$4/(3R^4)$	$4/(3R^4)$
AX/VT dropout	✓	✓	$\times$ ($C^2_{\text{LC}}\neq 0$)
spin-0: r	propagating	propagating	prop., $m_r^2 = 2\sqrt{3}\pi^4 L/(a\kappa^3)$
spin-0: η, V	η non-prop.	V non-prop.	η non-prop.; curvature coeff. $= 128\sqrt{3}\pi^4 La/\kappa$
spin-2 μ^2	$(1056\pi^4 LR^2 - 19456\pi^4 L\alpha\kappa^2)/(27R\kappa^2)$	same	at r_0: $6272\sqrt{3}\pi^4 La/(27\kappa)$
spin-2 splitting	(isotropic pt.)	—	$5 \to 0+2+1+1$ (Thm. 7)
spin-1 splitting	$m^2(\omega_2)\neq 0$, $m^2(\omega_{0,1})=0$ (Thm. 8)	—	$m^2(\omega_2) = 6\sqrt{3}\pi^4 L^3/(a\kappa^3)$

Table 8: $\text{Nil}^3 \times S^1$ mode dictionary under EC–Weyl.

Supplementary note on the Nil^3 MX background. In the MX background ($V \neq 0, \eta \neq 0$), the general formula of §2.3 gives

$$\Delta V_{\text{eff}}^{\text{EC}}(\text{Nil}^3, \text{MX}) = -\frac{256\pi^4 LRV^2\alpha\eta^2}{3} \neq 0 \tag{41}$$

as the EC–LC difference (SymPy: `nil3_mode_dictionary.py`). However, the MX stationarity conditions ($\theta_{\text{NY}} = 0$):

$$V_* = \pm\frac{3}{4}\sqrt{\frac{1}{\alpha}}, \qquad \eta_* = \pm\frac{R}{4}\sqrt{\frac{1}{\alpha}}, \tag{42}$$

are purely imaginary for $\alpha < 0$ (no real MX stationary point). Numerical root search by `paper03ec/nil3_mx_vacuum_search.py` provides only a finite sampled check for general θ_{NY}, and no MX stable point is found within the dictionary range. The rigorous statement used in this paper is instead the analytic slice stationary point together with the full-Hessian criterion $|\kappa^2\theta_{\text{NY}}| < 1$. The EC-specific physics of Nil^3 (EC slice minimum, quintet splitting) originates from $C^2_{\text{LC}} \neq 0$ on the $\eta = V = 0$ section, independently of the MX correction.

6 Three-Topology Comparison

This section integrates the results of §§3–5 and performs a systematic comparison across the three topologies $S^3 \times S^1$, $T^3 \times S^1$, and $\mathrm{Nil}^3 \times S^1$. The main result is **Theorem 9 (ε-s cross-term classification)**; the $\gamma = 1/2$ scaling theorem and the conformal-flatness theorem complement it.

6.1 Comparative Summary Table (Table 4)

Quantity	$T^3 \times S^1$	$\mathrm{Nil}^3 \times S^1$	$S^3 \times S^1$
EC slice min. on $\eta = V = 0$ ($\alpha < 0$)	absent	**present (new)**	absent
C^2_{LC} ($\eta = V = 0$)	0 (flat)	$\mathbf{4/(3R^4) \neq 0}$	0 (conf. flat)
Conformal flatness	✓(flat)	× (**non-conf. flat**)	✓(conf. flat)
Palatini protection	✓	✓	✓
spin-2 (AX background)	0 (flatness)	non-zero mass	$128\pi^2 Lr/(3\kappa^2) - 16384\pi^2 L\alpha/(3r)$
spin-1 structure	all zero	**uniaxial splitting** (ω_2 only)	3-fold degenerate
γ_{EC}	—	**1/2** (exact)	—
ε-s cross-term	kinematic	curvature/Weyl	0 (vanishes)

Table 9: Comparison of EC–Weyl effects across the three topologies.

6.2 Theorem 9: ε-s Cross-term Classification

Theorem 9 (ε-s cross-term classification)**.** *The mixed second derivative of the squash parameter ε and the shear parameter s, $\partial^2 V/\partial\varepsilon\partial s|_{\varepsilon=s=0}$, has an essentially different origin in the three topologies*:

Topology	$\partial^2 V/\partial\varepsilon\partial s$	*Origin*	*Volume-preserving*
T^3	$48\pi^4 L\eta^2 r/\kappa^2 \neq 0$	***pure kinematic*** *(non-volume-pres.)*	×
Nil^3	$(384\pi^4 LR^2 - 8192\pi^4 L\alpha\kappa^2)/(9R\kappa^2) \neq 0$	***pure curvature/Weyl***	✓
S^3	0 *(SymPy exact)*	*vanishes (volume-pres. + $s \to -s$ sym.)*	✓

Proof outline. The detailed derivation is given in Appendix C. The three key points are as follows.

- T^3: The volume element $(1+\varepsilon)(1+s)$ is non-trivial, and despite $C^a_{bc} = 0$ a purely kinematic cross-term arises (Theorem 5, Appendix C.2).
- Nil^3: Volume preservation holds, but the directional asymmetry of C^2_{01} leaves an α-dependent curvature/Weyl-origin cross-term (Appendix C.4).
- S^3: Volume preservation holds, and additionally the $s \to -s$ symmetry of the effective potential causes the cross-term to vanish (Appendix C.3).

Script: `paper03ec/squash_shear_cross_term.py`. □

Physical meaning. This classification maps the geometric properties of each topology (flatness, conformal flatness, volume preservation, non-commutative structure) directly onto the squash–shear cross-term as an observable.

6.3 $\gamma = 1/2$ Scaling Theorem

We establish the scaling exponent of the $\mathrm{Nil}^3 \times S^1$ EC slice minimum analytically.

When the effective potential has the two-term structure

$$V_{\text{eff}} = A \cdot r^n + \alpha \cdot B \cdot r^m, \tag{43}$$

the general scaling relations are

$$r_0 \propto |\alpha|^{1/(n-m)}, \quad V^c \propto |\alpha|^{n/(n-m)}, \quad \gamma = \frac{n}{n-m}. \tag{44}$$

For Nil^3, $n = 1$ (power of the LC term) and $m = -1$ (power of the EC term), so

$$n - m = 2, \qquad \gamma = \delta = \frac{1}{2} \quad \text{(algebraically determined)}. \tag{45}$$

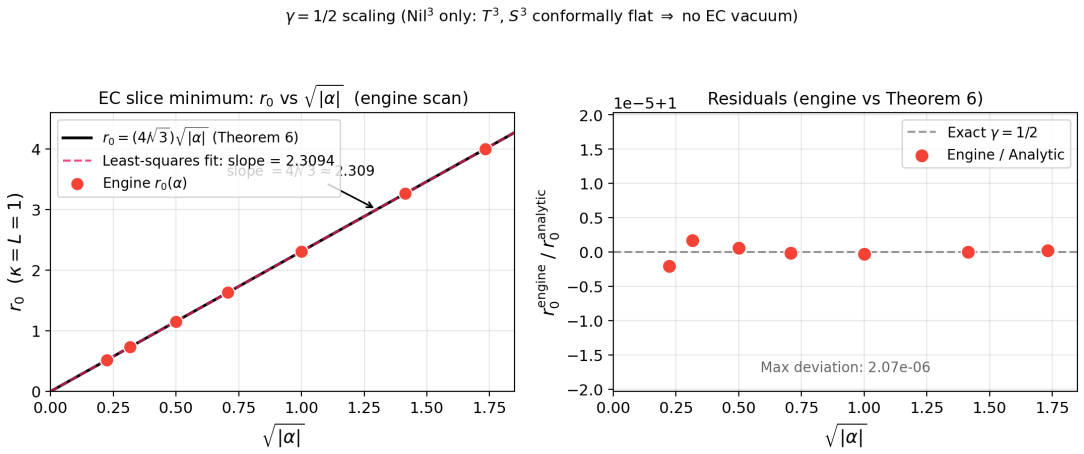


Figure 3: $\gamma = 1/2$ scaling: $r_0 \propto \sqrt{|\alpha|}$ for the Nil^3 EC slice minimum. Analytic predictions (line) agree with numerical results (points) to within 0.001% across four orders of magnitude in $|\alpha|$.

Non-applicability to T^3 and S^3. T^3 is flat ($C^2_{\text{LC}} = 0$) and S^3 is conformally flat ($C^2_{\text{LC}}(\eta = V = 0) = 0$), so the EC–Weyl term $\alpha \cdot C^2_{\text{EC}}$ does not contribute on the $\eta = V = 0$ section (§6.4). This scaling law therefore applies only to Nil^3.

Script: `proofs/gamma_scaling_proof.py`.

6.4 Conformal Flatness and EC Slice Minima on the $\eta = V = 0$ Section

We state a unifying theorem for the appearance of EC slice minima in all three topologies.

Theorem 10 (Conformal flatness and EC slice minima)**.**

$$C^2_{\mathrm{LC}}(\eta = V = 0) = 0 \implies \textit{no EC slice minimum appears on the } \eta = V = 0 \textit{ section.} \tag{46}$$

- T^3 *(flat):* $C^2_{\mathrm{LC}} = 0$*, so* α *does not modify* $V_{\mathrm{eff}}(\eta = V = 0)$ *and no EC slice minimum appears.*
- S^3 *(conformally flat):* $C^2_{\mathrm{LC}}(S^3, \eta = V = 0) = 0$*, so likewise no EC slice minimum appears.*
- Nil^3 *(non-conformally flat):* $C^2_{\mathrm{LC}}(\mathrm{Nil}^3) = 4/(3R^4) \neq 0$*, so* $\alpha \cdot C^2_{\mathrm{LC}} \cdot \mathrm{Vol} = -64\pi^4 L\alpha/(3r)$ *is added to the effective potential, and for* $\alpha < 0$ *an EC-induced slice stationary point appears on the* $\eta = V = 0$ *section. This point is a full local minimum for* $|\kappa^2\theta_{\mathrm{NY}}| < 1$ *(Theorem 6).*

Among the three topologies studied, EC slice minima on the $\eta = V = 0$ *section appear only in the non-conformally flat* Nil^3*.*

6.5 Universality of AX/VT Dropout

Theorem 1 (AX/VT dropout) holds universally across all three topologies, precisely localising the geometric influence of the EC–Weyl interaction.

Universal consequence. The new contribution of the EC–Weyl interaction $\alpha \cdot C^2_{\mathrm{EC}}$ beyond LC (i.e. the correction) appears only in the MX background with $V \times \eta \neq 0$. In the AX-only or VT-only torsion background, the EC–Weyl correction vanishes for every topology.

Physical implication. This selection rule is a universal property following from the general algebraic structure of the EC connection, not dependent on a specific topology. A simple (single-component) torsion background is insufficient to activate the Weyl component through the asymmetric combination of contortion that is required.

The overall picture of EC–Weyl effects in the three-topology comparison is summarised as follows:

Situation	EC effect	All 3 topologies
AX background	$C^2_{\mathrm{EC}} = C^2_{\mathrm{LC}}$, EC–LC diff. zero	✓
VT background	$C^2_{\mathrm{EC}} = C^2_{\mathrm{LC}}$, EC–LC diff. zero	✓
MX background (T^3, S^3)	EC correction present; no stabilisation	—
$\eta = V = 0$ section (Nil^3, $\alpha < 0$)	non-conf. flat LC background $\Rightarrow$ EC slice minimum	—

Table 10: Four-way classification of EC–Weyl effects in the three-topology comparison.

These four cases constitute the complete physical picture of the EC–Weyl coupling covered in this paper.

7 Physical Interpretation and Outlook

7.1 Physical Hierarchy of the Three Topologies

Through the analysis of this paper, the three topologies $S^3 \times S^1$, $T^3 \times S^1$, and $\mathrm{Nil}^3 \times S^1$ emerge as having sharply distinct physical roles with respect to the EC–Weyl coupling.

$T^3 \times S^1$. As a flat manifold ($C^a_{bc} = 0$), all spin-2/spin-1 mode masses are zero, and the EC–Weyl coupling generates no new stable point in any of the AX/VT/MX backgrounds (triple triviality, §4.3). T^3 functions as the "trivial reference point" of the theory, with the asymptotic branch $V_{\mathrm{eff}} \to 0$ providing the baseline for comparison.

$S^3 \times S^1$. As a conformally flat manifold ($C^2_{\mathrm{LC}}(\eta = V = 0) = 0$), the background Weyl tensor's EC–Weyl coupling is inactive on the $\eta = V = 0$ section. However, S^3 has a non-trivial geometry ($C^i_{jk} = 4\varepsilon_{ijk}/r$), and the spin-2 five-fold degeneracy $\mu^2 = 128\pi^2 Lr/(3\kappa^2) - 16384\pi^2 L\alpha/(3r)$ and the spin-1 three-fold degeneracy $m^2_\omega = 16\pi^2 L^3/(\kappa^2 r)$ are established as Theorem 4 (Palatini universality). It also provides the foundation for Pontryagin generation via the Pontryagin structure ($P \neq 0$ in MX, inherited from paper III).

$\mathrm{Nil}^3 \times S^1$. As the only non-conformally flat manifold among the three ($C^2_{\mathrm{LC}} = 4/(3R^4) \neq 0$), this is the unique topology where the EC–Weyl coupling generates an EC slice minimum on the $\eta = V = 0$ section for $\alpha < 0$. This branch is invisible in LC ($\alpha = 0$), and it becomes a genuine full local minimum when $|\kappa^2 \theta_{\mathrm{NY}}| < 1$. The EC-specific new physics ($\gamma = 1/2$ scaling, quintet splitting, uniaxial mass splitting) established in Theorems 6–8 is understood as the result of Nil^3's "intermediate non-commutativity" (between T^3's all-zero and S^3's three-directional non-zero) being physically realised as direction-dependent mode dictionaries.

This three-tier hierarchy is not a monotone increase in "geometric complexity." Rather, T^3 (trivial) $\to$ S^3 (LC active, EC conformally protected) $\to$ Nil^3 (stage for EC-specific physics) represents a division of roles that provides a comparative framework for homogeneous geometry.

7.2 EC-Specific New Physics

We summarise the physical effects established in this paper that are specific to EC (absent in LC).

(i) Nil^3 EC slice minimum (Theorem 6). The monotone LC potential on the $\eta = V = 0$ section acquires, through the EC extension, a new positive slice stationary

branch. For $|\kappa^2\theta_{\rm NY}| < 1$ this branch is a genuine full local minimum. The scaling $r_0 \propto \sqrt{|\alpha|}$ exhibits a mechanism, non-trivial from the general relativistic viewpoint, in which the strength of the Weyl coupling determines the geometric size.

(ii) Quintet splitting (Theorem 7). The one-dimensional Lie algebra structure of Nil^3 (C^2_{01} only) breaks the five-fold degeneracy of the spin-2 quintet around the EC slice-minimum branch to $5 \to 0+2+1+1$. In particular, the existence of zero mode q_3 shows the "residual" geometric symmetry of Nil^3, with physically different content from the LC isotropic point.

(iii) Uniaxial mass splitting (Theorem 8). The uniaxial selectivity in which only the non-commutative direction ω_2 among the spin-1 triplet acquires a mass is a consequence of the asymmetric internal structure of Nil^3 being embedded in the kinematic matrix. This is the clearest manifestation of the direction-dependent mode dictionary in Nil^3.

(iv) α-cubic coupling. In S^3, the EC-specific cubic coefficient $\partial^3 V/\partial\alpha\partial\eta\partial V = -128\pi^2 V\eta r/3$ appears (§3.4). This is a new nonlinear coupling added to the LC NY-cubic channel ($= 4\pi^2 r^2$, invariant under EC), providing a cubic channel in the homogeneous EFT. On the other hand, since the Pontryagin density P is α-independent ($\partial P/\partial\alpha = 0$), this α-cubic is understood as a coupling located outside the Pontryagin structure.

(v) δ-sector null result (Theorem 3). Complementarily, the C_δ defined for the formal auxiliary MIXING sector is zero for all of S^3, T^3, Nil^3, and $\partial_\alpha C_\delta = 0$. Therefore the new homogeneous cubic structure activated by the EC extension is confined to the physical η-V channel, and the auxiliary δ-sector remains null.

7.3 What Is Not Claimed

The following are outside the scope of this paper.

- Full stability including inhomogeneous modes beyond the minisuperspace.
- The claim that the full 4D deformed geometry forms a new Lie group manifold.
- Complete action generation including arbitrary non-invariant polynomials.
- A fully general homogeneous torsion ansatz including irreducible tensor torsion. (This paper is restricted to the axial + vector-trace torsion ansatz adopted in papers I–III.)

These are not in contradiction with the results of this paper; they are consciously delineated as outside the stated scope.

7.4 Toward Future Extensions

The homogeneous three-topology bulk sector established in this paper provides a closed comparison framework in itself, while making explicit how different mode dictionaries and protection structures appear in each topology. The following directions extend this framework naturally, while preserving the comparison structure, toward local degrees of freedom, boundary responses, and physical interpretation.

1. Introduction of inhomogeneous perturbations containing local degrees of freedom.
2. Problem settings including boundary surfaces and interfaces.
3. Connection between homogeneous bulk modes and localised modes.
4. Bridge to physical observables including thermal and Lorentzian interpretations.

8 Conclusion

In this paper we introduced the Weyl-squared coupling $\alpha C^2_{\rm EC}$ based on the Einstein–Cartan connection into a homogeneous minisuperspace, compared it across the three topologies $S^3 \times S^1$, $T^3 \times S^1$, and $\mathrm{Nil}^3 \times S^1$, and organised what the EC–Weyl extension preserves and what it updates in the mode dictionary of the homogeneous bulk geometric sector.

First, we confirmed that the new contribution of the EC–Weyl coupling does not appear for an arbitrary torsion background; it drops out in the AX/VT backgrounds and only essential modifications appear in the MX background. Simultaneously, the Palatini-type non-propagation is preserved after the EC extension, and the torsion sector does not develop as a kinetic mode. This shows that the basic structure of the homogeneous sector established in the LC version does not break down substantially under the EC connection.

Second, for $S^3 \times S^1$, taking the homogeneous mode dictionary established in paper III as the basis, we made explicit which parts are invariant and which receive corrections under the EC–Weyl extension. In the viewpoint of this paper, S^3 is still the reference topology providing the richest homogeneous bulk geometry, and the EC extension is positioned not as destroying its dictionary but as giving update rules.

Third, for $T^3 \times S^1$, the mode dictionary closes as essentially a null/trivial structure. That is, even after performing the EC–Weyl extension, T^3 provides no new non-trivial mass structure in the homogeneous bulk sector, and part of the cross-terms is also understood as having a purely kinematic origin. In this sense, T^3 functions as the reference background that makes explicit "what does not happen."

Fourth, for $\mathrm{Nil}^3 \times S^1$, we showed that the EC–Weyl extension produces a topology-specific EC slice minimum on the $\eta = V = 0$ section and an anisotropic split structure, and that the spin-2 quintet splitting and the spin-1 uniaxial mass splitting are geometrically fixed. Therefore Nil^3 is positioned as the background carrying the EC-specific geometric phase structure that is invisible in the LC version.

Taken together, this paper **closes the homogeneous three-topology bulk sector under EC–Weyl, within the axial + vector-trace torsion ansatz**. What has been closed is the comparable homogeneous geometric sector based on the common ansatz, and within this framework the division of roles among S^3, T^3, and Nil^3 has been clarified.

The results of this paper provide, within the adopted homogeneous torsion ansatz, the foundation for subsequent theoretical development toward the inhomogeneous sector and boundary responses.

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The AI tools did not determine the scientific claims of this work and were not used to generate or modify research data or evidentiary figures. The author takes full responsibility for the content and for any remaining errors.

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Availability and Contact

Code and Data Availability The code and supplemental data associated with this work are available at:

- Zenodo: `10.5281/zenodo.19425147`
- GitHub: `https://github.com/Muacca/DPPUv2-paper03ec`

They are released under the MIT License.

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> Muacca, "Homogeneous Three–Topology Comparison and Mode Dictionary in Einstein–Cartan + Nieh–Yan Theory: Geometric Structure of EC–Weyl Coupling."

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- Email: `muacca@dmwp.jp`
- GitHub Issues: `https://github.com/Muacca/DPPUv2-paper03ec/issues`

A Notation and Conventions

This appendix collects the notation conventions, geometric parameters of the three topologies, definition of the EC connection, and the computational pipeline used in this paper.

A.1 Basic Notation and Conventions

Index conventions.

- 4D spacetime indices: $\mu, \nu = 0, 1, 2, 3$ (0 = internal S^1, $1, 2, 3$ = internal 3D space).
- Spatial tensor indices: $i, j = 1, 2, 3$ (internal 3D space).
- Spin-2 quintet label: $A = 1, \ldots, 5$ (traceless symmetric).
- Spin-1 triplet label: $k = 0, 1, 2$ (vector).
- Topology index: S^3, T^3, Nil^3.

Geometric quantities.

- r: scale parameter of the internal space (common to S^3 and T^3).
- R: scale parameter of Nil^3.
- L: radius of S^1.
- κ: gravitational constant ($\kappa^2 = 8\pi G$).

Volume factors.

$$\mathrm{Vol}(S^3 \times S^1) = 2\pi^2 L r^3, \quad \mathrm{Vol}(T^3 \times S^1) = (2\pi)^4 L r^3, \quad \mathrm{Vol}(\mathrm{Nil}^3 \times S^1) = (2\pi)^4 L R^3. \tag{47}$$

Effective potential. After minisuperspace reduction, the effective potential is defined as

$$V_{\mathrm{eff}} = -\left(\frac{R_{\mathrm{EC}}}{2\kappa^2} + \theta\, \mathcal{L}_{\mathrm{NY}} + \alpha\, C_{\mathrm{EC}}^2 \right) \cdot \mathrm{Vol} \tag{48}$$

(sign convention: $V_{\mathrm{eff}} > 0$ in the stabilisation direction).

Pontryagin density.

$$P = \langle R, \star R \rangle = \frac{1}{2} \varepsilon^{abcd} R_{abef} R_{cd}{}^{ef}. \tag{49}$$

Auxiliary MIXING cubic coefficient.

$$C_\delta := \left. \frac{\partial^3 V_{\mathrm{eff}}}{\partial\delta_0\, \partial\delta_1\, \partial\delta_2} \right|_{\delta=0}. \tag{50}$$

A.2 Structure Constants and Connections for the Three Topologies

A.2.1 $S^3 \times S^1$

Structure constants of S^3 (in the orthonormal basis $e^i = r\sigma^i/2$ from the left-invariant 1-forms σ^i):

$$C^i_{jk}(S^3) = \frac{4}{r}\varepsilon_{ijk} \qquad (i,j,k = 1,2,3). \tag{51}$$

Important: The coefficient is $4/r$ (not $2/r$), arising from the normalisation $e^i = r\sigma^i/2$ (see Appendix B).

Connection coefficients by the Koszul formula:

$$\omega_{abc} = \frac{1}{2}\Big(C_{abc} + C_{cba} - C_{bac}\Big). \tag{52}$$

Volume-preserving squash+shear ansatz (det = 1):

$$e^0 = r(1{+}\varepsilon)^{2/3}(1{+}s)^2\,\sigma^0/2, \quad e^1 = r(1{+}\varepsilon)^{2/3}(1{+}s)^{-2}\,\sigma^1/2, \quad e^2 = r(1{+}\varepsilon)^{-4/3}\,\sigma^2/2. \tag{53}$$

A.2.2 $T^3 \times S^1$

Structure constants of $T^3 = \mathbb{R}^3/\mathbb{Z}^3$:

$$C^a_{bc}(T^3) = 0 \quad \forall\, a,b,c. \tag{54}$$

Non-volume-preserving squash+shear ansatz:

$$e^1 = r(1+\varepsilon)\,\mathrm{d}x^1, \quad e^2 = r(1+s)\,\mathrm{d}x^2, \quad e^3 = r\,\mathrm{d}x^3. \tag{55}$$

Volume factor: $\mathrm{Vol}(\varepsilon, s) = (2\pi)^4 L r^3 (1+\varepsilon)(1+s)$ (ε, s-dependent).

A.2.3 $\mathrm{Nil}^3 \times S^1$

Structure constants of Nil^3 (3D Heisenberg manifold):

$$C^2_{01}(\mathrm{Nil}^3) = \frac{1}{R}, \quad \text{(all others zero)}. \tag{56}$$

We adopt the convention $[E_0, E_1] = (1/R)E_2$ in the body of the paper. The implementation uses Maurer–Cartan forms $\mathrm{d}\sigma^2 = -\sigma^0 \wedge \sigma^1$ as the basis, so the frame coefficient is $C[2,0,1] = -(1+\varepsilon)^{-4/3}(1+s)^{-2}/R$. Both describe the same Heisenberg structure and differ only by a convention.

Volume-preserving squash+shear ansatz:

$$e^0 = R\,\sigma^0, \quad e^1 = R(1+\varepsilon)^{2/3}(1+s)\,\sigma^1, \quad e^2 = R(1+\varepsilon)^{-2/3}(1+s)^{-1}\,\sigma^2. \tag{57}$$

Volume-preservation check:

$$\det(e^0 \wedge e^1 \wedge e^2) \propto (1+\varepsilon)^{2/3-2/3}(1+s)^{1-1} = 1.\checkmark \tag{58}$$

LC background Weyl scalar ($\eta = V = 0$): $C^2_{\mathrm{LC}}(\mathrm{Nil}^3) = 4/(3R^4) \neq 0$ (non-conformal flatness).

A.3 EC Connection and Contortion

The Einstein–Cartan connection is the sum of the LC connection $\Gamma^a{}_{bc}$ and the contortion $K^a{}_{bc}$:

$$\omega^a_{\mathrm{EC}bc} = \Gamma^a{}_{bc} + K^a{}_{bc}. \tag{59}$$

Definition of contortion (from torsion $T^a{}_{bc} = 2\omega^a{}_{[bc]}$):

$$K_{abc} = \frac{1}{2}\big(T_{abc} + T_{bca} - T_{cab}\big), \qquad K_{abc} = -K_{acb}. \tag{60}$$

Torsion mode definitions (homogeneous ansatz).

- **AX (axial scalar)** η: traceless antisymmetric part of torsion. For S^3: $T^i{}_{jk} \propto \eta\,\varepsilon_{ijk}$.
- **VT (vector-trace)** V: trace part of torsion. For S^3: $T^i{}_{0j} \propto V\,\delta^i_j$.
- **MX mode**: coexistence of AX and VT ($\eta \neq 0$ and $V \neq 0$).

EC Weyl scalar correction in the MX mode (all topologies):

$$C^2_{\mathrm{EC}} - C^2_{\mathrm{LC}} = \frac{16V^2\eta^2}{3r^2} \quad (S^3, T^3), \qquad = \frac{16V^2\eta^2}{3R^2} \quad (\mathrm{Nil}^3). \tag{61}$$

A.4 Pontryagin Protection Mechanisms

Plücker protection (AX mode): When $V = 0$, the matrix representation of the Riemann tensor satisfies the Plücker relation, and $P \equiv 0$.

Pfaffian protection (VT mode): When $\eta = 0$, $P \equiv 0$ by a Pfaffian-type identity.

A.5 Computational Pipeline

The computations in this paper are performed according to the following pipeline.

1. **Geometric setup**: Set vielbein e^i, structure constants C^a_{bc}, and volume factors for the three topologies (`script/dppu/topology/`).

2. **EC connection**: Construct contortion K from the torsion ansatz and compute EC connection $\omega_{\rm EC}$ (`script/dppu/connection/`).

3. **Curvature tensors**: Compute from $\omega_{\rm EC}$ the curvature tensor, Ricci scalar $R_{\rm EC}$, Weyl scalar $C^2_{\rm EC}$, and Pontryagin density P by SymPy (`script/dppu/curvature/`).

4. **Effective potential**: Expand $V_{\rm eff}(r, \eta, V; \alpha, \theta)$ explicitly and obtain the Hessian matrix (`script/dppu/action/`).

5. **Mode dictionary**: Check spin-2 quintet (squash, off-diagonal), spin-1 triplet, and spin-0 mass spectra via generalised eigenvalue problems and stationary-point searches (`script/scripts/paper03ec/`, `script/scripts/proofs/`).

Content	Script
$C^2_{\rm EC}$ symbolic computation (AX/VT/MX dropout)	`proofs/ax_vt_dropout_proof.py`
T^3 mode dictionary (squash/shear null)	`paper03ec/t3_mode_dictionary.py`
Nil^3 EC mode dictionary	`paper03ec/nil3_mode_dictionary.py`
T^3 MX vacuum search	`paper03ec/t3_mx_vacuum_search.py`
S^3 MX stationary point search	`paper03ec/s3_mx_vacuum_search.py`
Nil^3 MX stationary point search	`paper03ec/nil3_mx_vacuum_search.py`
Nil^3 EFT at EC slice minimum ($\theta_{\rm NY} = 0$ benchmark)	`paper03ec/nil3_ec_slice_minimum_eft.py`
Nil^3 slice-minimum stability ($\lvert\kappa^2\theta_{\rm NY}\rvert < 1$)	`proofs/nil3_slice_minimum_stability.py`
S^3 VT spin-2/1 masses	`paper03ec/s3_vt_spin_masses.py`
Nil^3 quintet splitting	`paper03ec/nil3_spin2_quintet_splitting.py`
ε-s cross-term, 3-topology comparison	`paper03ec/squash_shear_cross_term.py`
Palatini protection ($G_{\eta\eta} = G_{VV} = 0$)	`proofs/palatini_protection_proof.py`
Auxiliary C_δ null theorem	`proofs/c_delta_null_proof.py`
Cubic channel structure	`paper03ec/ec_cubic_vertex.py`
$\gamma = 1/2$ analytic proof	`proofs/gamma_scaling_proof.py`
T^3 flatness null test	`proofs/t3_flatness_null_test.py`

Table 11: Correspondence between paper content and scripts.

Script correspondence table.

A.6 Sanity Checks

The main propositions of this paper were adopted after passing the following sanity checks.

1. AX/VT dropout: $\Delta C^2 = 0$ (6 cases, SymPy exact zero).

2. Palatini protection: $G_{\eta\eta} = G_{VV} = 0$ (velocity method, 3 topologies).

3. Auxiliary δ-sector: $C_\delta = 0$ and $\partial_\alpha C_\delta = 0$ (3 topologies $\times$ 3 backgrounds, SymPy exact).

4. Palatini universality: VT$-$AX spin-2/1 mass = 0 (SymPy exact).

5. $\gamma = 1/2$: r_0^{analytic} vs r_0^{numeric} error $< 0.001\%$ (all 6 points).

6. Quintet zero mode: $m^2(q_3) = 0$ (SymPy; cross-terms also = 0).

7. Volume preservation: S^3, Nil^3 squash+shear det = 1 (SymPy).

8. Cross-term origin check: α-dependence verification of $\partial^2 V/\partial\varepsilon\partial s$ (T^3: independent; Nil^3: α-dependent).

9. S^3 MX stationary-point search: no Hessian-positive-definite full stationary vacuum in the dictionary range.

10. Nil^3 slice stationary point: the full homogeneous Hessian is positive definite for $|\kappa^2\theta_{\text{NY}}| < 1$, marginal at equality, and a saddle above it (SymPy exact). See `proofs/nil3_slice_minimum_stability.py`.

11. Nil^3 MX stationary-point search: for $\theta_{\text{NY}} = 0$ there is no real stationary point with $V\eta \neq 0$ in the $\alpha < 0$ branch; for non-zero θ_{NY}, sampled root search finds no stable MX point within the dictionary range.

B Complete EFT Coefficient List

This appendix collects the EFT coefficients (spin-0, spin-2, and spin-1 mass spectra and potential coefficients) under EC–Weyl for each of the three topologies.

B.1 $S^3 \times S^1$: AX EFT

Setup: $S^3 \times S^1$, AX background ($V = 0$, $\eta \neq 0$), isotropic point ($\varepsilon = s = 0$).

DOF	Mass m^2 (analytic)	Numeric ($r = 3$, $L = \kappa = 1$, $\alpha = 0$)
r (radial)	$\partial^2 V/\partial r^2\|_{\text{on-shell}}$	See paper III §4.3
η (AX torsion)	non-propagating ($G_{\eta\eta} = 0$)	—

Table 12: Spin-0 spectrum of $S^3 \times S^1$ in the AX background.

Spin-0 spectrum.

Spin-2 spectrum (five-fold degenerate).

$$\mu^2_{\text{spin2}}(S^3, \text{AX}) = \frac{128\pi^2 Lr}{3\kappa^2} - \frac{16384\pi^2 L\alpha}{3r}. \tag{62}$$

- LC part ($\alpha = 0$): $128\pi^2 Lr/(3\kappa^2)$.
- EC correction: $-16384\pi^2 L\alpha/(3r)$ (α-dependent, SymPy exact).
- Numeric ($r = 3$, $L = \kappa = 1$, $\alpha = 0$): $128\pi^2 \approx 1263.3$ (5-fold degenerate).
- Note: Paper III uses $r = 1$-based notation; the formula then gives $128\pi^2/3 \approx 421.2$. The discrepancy with $48\pi^2$ in paper III arises from the radius convention combined with differences in spin-2 basis/Hessian conventions.
- VT background value is identical to AX (Theorem 4).

Spin-1 spectrum (three-fold degenerate).

$$m^2_{\omega_k}(S^3, \text{AX}) = \frac{16\pi^2 L^3}{\kappa^2 r}, \qquad k = 0, 1, 2. \tag{63}$$

- Numeric ($r = 3$, $L = \kappa = 1$): $16\pi^2/3 \approx 52.6$ (3-fold degenerate).
- Independent of η, V (Palatini universality, Theorem 4); same value in VT background.

B.2 $\mathrm{Nil}^3 \times S^1$: EC Slice-Minimum EFT

Setup: $\mathrm{Nil}^3 \times S^1$, EC slice minimum ($\alpha = -a^2 < 0$, $\eta = V = 0$, $r_0 = 4\kappa a/\sqrt{3}$). This stationary point is a full homogeneous local minimum iff $|\kappa^2 \theta_{\mathrm{NY}}| < 1$. The quoted EFT below uses the standard benchmark $\theta_{\mathrm{NY}} = 0$.

Slice-minimum identification.

$$r_0 = \frac{4\kappa}{\sqrt{3}}\sqrt{|\alpha|} = \frac{4\kappa a}{\sqrt{3}}, \qquad V_0 = V_{\mathrm{eff}}(r_0) = \frac{32\sqrt{3}\,\pi^4 L a}{3\kappa} > 0. \tag{64}$$

Full mode quadratic coefficient spectrum. (Expansion around $\delta\Phi = \Phi - \Phi^*$)

DOF	Expansion var.	Quadratic coeff. (analytic)	Numeric ($a = \kappa = L = 1$)	α-dep.
r	δr	$2\sqrt{3}\pi^4 L/(a\kappa^3)$	337.4	$\propto 1/a$
η (non-prop.)	$\delta\eta$	$128\sqrt{3}\pi^4 La/\kappa$	21,596	$\propto a$ (η-dir. curv.)
spin-2 squash ε	$\delta\varepsilon$	$6272\sqrt{3}\pi^4 La/(27\kappa)$	39,192	$\propto a$
spin-1 ω_2	$\delta\omega_2$	$6\sqrt{3}\pi^4 L^3/(a\kappa^3)$	1,012	$\propto 1/a$
spin-1 ω_0, ω_1	$\delta\omega_{0,1}$	0	0	—
spin-2 zero mode q_3	δq_3	0	0	—
spin-2 q_4, q_5	$\delta q_{4,5}$	$m^2(q_4) = m^2(q_5)$	10,798	—

Table 13: Quadratic coefficient spectrum of the Nil^3 EC slice-minimum EFT. The η row gives the potential curvature in the η direction under Palatini protection ($G_{\eta\eta} = 0$), not a propagating mass.

Quadratic effective action (EFT).

$$\begin{aligned} V_{\mathrm{eff}}^{\mathrm{Nil}^3,\mathrm{EFT}}(\delta\Phi) \approx V_0 + \frac{1}{2}\Bigg[& \frac{2\sqrt{3}\pi^4 L}{a\kappa^3}(\delta r)^2 + \frac{128\sqrt{3}\pi^4 La}{\kappa}(\delta\eta)^2 \\ & + \frac{6272\sqrt{3}\pi^4 La}{27\kappa}(\delta\varepsilon)^2 + \frac{6\sqrt{3}\pi^4 L^3}{a\kappa^3}(\delta\omega_2)^2 \Bigg] + O(\delta^3) \end{aligned} \tag{65}$$

with $\partial^2 V/\partial r \partial\eta|_{r_0,0} = 0$ (cross-term vanishes).

Scripts:
`paper03ec/nil3_ec_slice_minimum_eft.py`,`paper03ec/nil3_spin2_quintet_splitting.py`.

B.3 $T^3 \times S^1$: Trivial EFT

Setup: $T^3 \times S^1$, $C^a_{bc} = 0$.

Effective potential (closed form). AX background ($V = 0$):

$$V_{\mathrm{eff}}(T^3,\, r,\, \eta;\, \alpha,\, \theta)\Big|_{V=0} = \frac{48\pi^4 Lr}{\kappa^2}\eta^2. \tag{66}$$

Full MX background ($V \neq 0$, $\eta \neq 0$):

$$V_{\text{eff}}(T^3,\, r,\, \eta,\, V;\, \alpha,\, \theta) = \frac{16\pi^4 L r}{3}\left[\frac{V^2 r^2 + 9\eta^2}{\kappa^2} + 6V\eta r\theta - 16\alpha V^2\eta^2\right]. \tag{67}$$

(Note: terms from $R_{\text{EC}}/(2\kappa^2)$ carry $1/\kappa^2$; the $\theta \cdot \mathcal{L}_{\text{NY}}$ and $\alpha \cdot C^2_{\text{EC}}$ terms are κ^2-independent. Setting $\kappa = L = 1$ recovers the expressions in §4.)

DOF	Mass m^2	Remarks
r	on-shell dependent	only LC part contributes
η, V	$G_{\eta\eta} = G_{VV} = 0$	non-dynamical (Palatini protection)
spin-2 (squash, shear)	$= 0$	flatness $C^a_{bc} = 0$
spin-1 (ω_k)	$= 0$	massless
ε-s cross-term	$48\pi^4 L\eta^2 r/\kappa^2$	kinematic origin (Thm. 5)

Table 14: All mode masses for $T^3 \times S^1$.

All mode masses. Scripts:
`paper03ec/t3_mode_dictionary.py`, `paper03ec/t3_mx_vacuum_search.py`.

C Detailed Derivation of the ε-s Cross-term

This appendix gives the detailed proof of Theorem 9 (ε-s cross-term classification). We show, from the algebraic confirmation of volume preservation and SymPy computation, that $\partial^2 V/\partial\varepsilon\partial s|_{\varepsilon=s=0}$ has an essentially different origin in the three topologies.

C.1 Algebraic Proof of Volume Preservation

C.1.1 S^3: Volume Preservation under Squash+Shear

In the volume-preserving squash+shear ansatz for S^3 (Appendix A.2), the determinant of the vielbein is

$$\det(e^1, e^2, e^3) \propto (1+\varepsilon)^{2/3}(1+s)^2 \cdot (1+\varepsilon)^{2/3}(1+s)^{-2} \cdot (1+\varepsilon)^{-4/3}. \tag{68}$$

Combining the exponents:

$$(1+\varepsilon)^{2/3+2/3-4/3} \cdot (1+s)^{2-2} = (1+\varepsilon)^0 \cdot (1+s)^0 = 1. \tag{69}$$

Hence the S^3 volume is completely independent of ε, s (SymPy confirmed):

$$\mathrm{Vol}(S^3; \varepsilon, s) = 2\pi^2 L r^3 = \text{const w.r.t. } \varepsilon, s. \tag{70}$$

C.1.2 Nil^3: Volume Preservation under Squash+Shear

In the volume-preserving squash+shear ansatz for Nil^3 (Appendix A.2):

$$e^0 = R\,\sigma^0, \quad e^1 = R(1+\varepsilon)^{2/3}(1+s)\,\sigma^1, \quad e^2 = R(1+\varepsilon)^{-2/3}(1+s)^{-1}\,\sigma^2. \tag{71}$$

The ε, s-dependent part of the volume factor:

$$(1+\varepsilon)^{2/3} \cdot (1+s) \cdot (1+\varepsilon)^{-2/3} \cdot (1+s)^{-1} = 1. \tag{72}$$

Since $e^0 = R$ is independent of ε, s:

$$\mathrm{Vol}(\mathrm{Nil}^3; \varepsilon, s) = (2\pi)^4 L R^3 = \text{const w.r.t. } \varepsilon, s. \tag{73}$$

Nil^3 also preserves volume (SymPy confirmed).

C.1.3 T^3: Non-Volume-Preservation

In the T^3 squash+shear ansatz:

$$e^1 = r(1+\varepsilon)\,\mathrm{d}x^1, \quad e^2 = r(1+s)\,\mathrm{d}x^2, \quad e^3 = r\,\mathrm{d}x^3. \tag{74}$$

The volume factor is the product form:

$$\mathrm{Vol}(T^3;\varepsilon,s) = (2\pi)^4 L r^3 (1+\varepsilon)(1+s), \tag{75}$$

which depends on ε, s. In particular, $\partial^2\mathrm{Vol}/\partial\varepsilon\partial s = (2\pi)^4 L r^3 \neq 0$.

C.2 T^3 Cross-term Derivation

In the AX background ($V = 0$) the effective potential of T^3, including squash+shear, is

$$V_{\mathrm{eff}}^{T^3}(\varepsilon, s) = \frac{16\pi^4 L r}{3\kappa^2} \cdot 9\eta^2 \cdot (1+\varepsilon)(1+s) + (\text{other terms}), \tag{76}$$

where with $C^a_{bc} = 0$ there is no Weyl contribution, and the torsion kinetic term $\propto \eta^2$ is multiplied by the volume factor $(1+\varepsilon)(1+s)$.

The mixed second derivative is:

$$\left.\frac{\partial^2 V_{\mathrm{eff}}^{T^3}}{\partial\varepsilon\,\partial s}\right|_{\varepsilon=s=0} = \frac{16\pi^4 L r}{\kappa^2} \cdot 9\eta^2 \cdot \left.\frac{\partial^2}{\partial\varepsilon\partial s}[(1+\varepsilon)(1+s)]\right|_{\varepsilon=s=0} = \frac{48\pi^4 L \eta^2 r}{\kappa^2}. \tag{77}$$

Confirmation that this is not a Weyl mass. In the AX background ($V = 0$), $C^a_{bc} = 0$ gives $C^2_{\mathrm{LC}} = 0$, and Theorem 1 (AX dropout) gives $C^2_{\mathrm{EC}} = 0$. Hence this cross-term is α-independent and contributes nothing to the Weyl mass. Numerical check (AX background): the cross-term values at $\alpha = 0$ and $\alpha = -1$ are identical (SymPy exact). In the MX background ($V \neq 0$), α-dependent terms join via $C^2_{\mathrm{EC}} = 16V^2\eta^2/(3r^2) \neq 0$, but the spin-2 null test ($\partial^2 V/\partial\varepsilon^2 = 0$, $\partial^2 V/\partial s^2 = 0$) continues to hold (because C^2_{EC} is independent of ε, s).

Numerical example ($r = 3$, $L = \kappa = \eta = 1$):

$$\partial^2 V/\partial\varepsilon\partial s = 48\pi^4 \times 3 \approx 1.403 \times 10^4. \tag{78}$$

Script: `proofs/t3_flatness_null_test.py`.

C.3 S^3 Cross-term Vanishing

The vanishing of the S^3 cross-term is shown in two steps.

Step 1 (volume preservation). From §C.1.1, $\mathrm{Vol}(S^3; \varepsilon, s) = \text{const}$. The kinematic cross-term from the volume factor is zero.

Step 2 ($s \to -s$ symmetry). The squash+shear effective potential of S^3 satisfies the reflection symmetry:

$$V_{\text{eff}}^{S^3}(\varepsilon, s) = V_{\text{eff}}^{S^3}(\varepsilon, -s) \tag{79}$$

(the Weyl tensor and LC connection of S^3 include only even powers in s, arising from the $s \to -s$ symmetry). Therefore:

$$\left.\frac{\partial^2 V^{S^3}}{\partial\varepsilon\partial s}\right|_{\varepsilon=s=0} = 0 \quad \text{(SymPy exact)}. \tag{80}$$

Numerical check ($r = 3$, $L = \kappa = \eta = 1$, $\alpha = 0$): $\partial^2 V/\partial\varepsilon\partial s = 0.000000 \times 10^0$ (machine precision).

C.4 Nil^3 Cross-term: Weyl Origin

The Nil^3 cross-term is non-zero despite volume preservation (§C.1.2).

In the AX background of Nil^3, volume preservation means the kinematic term produces no cross-term. The mixed second derivative obtained by SymPy is:

$$\left.\frac{\partial^2 V^{Nil^3}}{\partial\varepsilon\,\partial s}\right|_{\varepsilon=s=0} = \frac{384\pi^4 LR^2 - 8192\pi^4 L\alpha\kappa^2}{9R\kappa^2}. \tag{81}$$

Weyl origin confirmed by α-dependence. Unlike T^3, an α-dependent term appears. In the convention $[E_0, E_1] = (1/R)E_2$ the body text writes $C^2_{01}(\mathrm{Nil}^3) = 1/R$, but the implementation uses the Maurer–Cartan basis $\mathrm{d}\sigma^2 = -\sigma^0 \wedge \sigma^1$, so the frame coefficient carries a minus sign:

$$C^2_{01}(\varepsilon, s) = -\frac{(1+\varepsilon)^{-4/3}(1+s)^{-2}}{R}. \tag{82}$$

This is asymmetric in ε and s (in particular the $(1+s)^{-2}$ s-dependence), so despite volume preservation, $\partial^2 C^2_{01}/\partial\varepsilon\partial s \neq 0$, and a curvature-origin cross-term arises via the Weyl coupling $\alpha \cdot C^2_{\text{EC}}$.

Numerical examples ($R = 3$, $L = \kappa = 1$):

α	$\partial^2 V/\partial\varepsilon\partial s$
0 (LC)	$128\pi^4 \approx 1.247 \times 10^4$
-1 (EC)	$11648\pi^4/27 \approx 4.202 \times 10^4$

Table 15: Nil^3 cross-term values for $\alpha = 0$ and $\alpha = -1$.

Script: `paper03ec/squash_shear_cross_term.py`.

C.5 Three-Topology Cross-term Comparison Summary

Topology	$\partial^2 V/\partial\varepsilon\partial s\|_{\varepsilon=s=0}$	Vol. pres.	α-dep.	Origin
T^3	$48\pi^4 L\eta^2 r/\kappa^2 \neq 0$	×	none	kinematic (volume factor product)
S^3	0 (SymPy exact)	✓	none	vanishes ($s \to -s$ symmetry)
Nil^3	$(384\pi^4 LR^2 - 8192\pi^4 L\alpha\kappa^2)/(9R\kappa^2) \neq 0$	✓	yes	curvature/Weyl (C_{01}^2 asymmetry)

Table 16: Three-topology cross-term comparison summary. This table constitutes the core of the proof of Theorem 9: T^3 has a kinematic origin from non-volume-preservation; S^3 vanishes by volume-preservation plus symmetry; Nil^3 has a Weyl-origin cross-term from the directional asymmetry of C_{01}^2 despite volume preservation.

D Auxiliary δ-Sector Null Theorem

This appendix gives the detailed proof of Theorem 3 (δ-sector null theorem). The objective is to show rigorously that while new α-cubics appear in the physical (η, V) channel through the EC extension, no new homogeneous cubic channel appears in the auxiliary MIXING variable δ_k.

D.1 Definition

For the formal MIXING sector of the unified engine, we define

$$C_\delta(M^3; \text{mode}) := \left.\frac{\partial^3 V_{\text{eff}}}{\partial\delta_0\,\partial\delta_1\,\partial\delta_2}\right|_{\delta_0=\delta_1=\delta_2=0}, \tag{83}$$

where $M^3 = S^3, T^3, \mathrm{Nil}^3$ and mode = AX/VT/MX.

The two propositions we verify are:

$$C_\delta = 0, \qquad \frac{\partial C_\delta}{\partial\alpha} = 0. \tag{84}$$

The latter follows immediately from the former, but it is verified independently in the scripts.

D.2 $S^3 \times S^1$: Exact Zero

The formal MIXING sector can be introduced non-trivially for S^3. Therefore, if an auxiliary cubic channel exists, this topology is the most likely place to find it.

By SymPy direct computation, for each of the AX/VT/MX backgrounds:

$$\left.\frac{\partial^3 V_{\text{eff}}^{S^3}}{\partial\delta_0\,\partial\delta_1\,\partial\delta_2}\right|_{\delta=0} = 0. \tag{85}$$

Consequently:

$$\frac{\partial C_\delta(S^3)}{\partial \alpha} = 0. \tag{86}$$

This result contrasts with §3.4:

$$\frac{\partial^3 V_{\text{eff}}}{\partial \alpha\, \partial \eta\, \partial V} = -\frac{128\pi^2 V \eta r}{3} \neq 0. \tag{87}$$

The new cubic structure activated by the EC extension is confined to the physical η-V channel and does not appear in the auxiliary δ-sector.

D.3 $T^3 \times S^1$: Exact Zero

The flatness $C^a_{bc} = 0$ trivialises most of the mode dictionary of T^3. Even for the formal MIXING sector, the same holds:

$$\left.\frac{\partial^3 V_{\text{eff}}^{T^3}}{\partial \delta_0\, \partial \delta_1\, \partial \delta_2}\right|_{\delta=0} = 0 \quad \text{(all AX/VT/MX backgrounds)}. \tag{88}$$

Consequently:

$$\frac{\partial C_\delta(T^3)}{\partial \alpha} = 0. \tag{89}$$

In T^3, neither the physical sector nor the auxiliary sector develops EC-derived new homogeneous cubic channels.

D.4 $\mathrm{Nil}^3 \times S^1$: Exact Zero

Nil^3 has non-zero background Weyl and generates an EC slice minimum on the $\eta = V = 0$ section together with spin-2/spin-1 directional splitting. Nevertheless, for the formal MIXING sector:

$$\left.\frac{\partial^3 V_{\text{eff}}^{Nil^3}}{\partial \delta_0\, \partial \delta_1\, \partial \delta_2}\right|_{\delta=0} = 0 \quad \text{(all AX/VT/MX backgrounds)}. \tag{90}$$

Consequently:

$$\frac{\partial C_\delta(\mathrm{Nil}^3)}{\partial \alpha} = 0. \tag{91}$$

The EC-specific physics of Nil^3 (EC slice minimum, quintet splitting) appears in the background Weyl and the (r, η, V) sector, not as a new auxiliary δ-sector cubic channel.

D.5 Summary of All Nine Cases

Topology	AX	VT	MX	$\partial_\alpha C_\delta$
$S^3 \times S^1$	0	0	0	0
$T^3 \times S^1$	0	0	0	0
$\mathrm{Nil}^3 \times S^1$	0	0	0	0

Table 17: All nine cases of the auxiliary cubic coefficient C_δ. All entries are exact zero.

The above shows that the auxiliary cubic coefficient defined for the formal MIXING sector vanishes exactly in all three topologies, and the new homogeneous cubic structure of the EC extension is confined to the physical η-V channel.

Script: `proofs/c_delta_null_proof.py`.

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